Free running CFD simulations to investigate ship manoeuvrability in waves

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ABSTRACT

Ship manoeuvrability is closely related to the safety of ship operation in a real seaway; therefore predicting a ship's manoeuvring characteristics is of great importance. However, estimating a ship's manoeuvring performance in waves is challenging because of the complex fluid-structure interactions between hull, rudder, propeller and waves. Traditional methods for the study of a ship's manoeuvrability in waves, based on mathematical models combined with potential flow theory, cannot accurately and effectively incorporate the effect of waves on a ship. On the other hand, unsteady Reynolds-Averaged Navier-Stokes (RANS) computations are capable of resolving the interactions between a ship and waves with high accuracy. The main objective of this study is to carry out a fully nonlinear unsteady RANS simulation to predict the manoeuvring behaviour of a KRISO Container ship (KCS) model in different wave conditions. A dynamic overset grid method was used with the purpose of resolving large ship motions and rudder movement. A course keeping and manoeuvring control module based on a feedback control system was embedded in the simulation to control rudder deflection according to standard manoeuvre mechanisms. Simulation results were compared with available experimental data and showed good agreement, which demonstrates that CFD is a reliable approach to estimate ship manoeuvrability in waves.

1. Introduction

Understanding a ship manoeuvring performance in a real seaway is of great importance since it is intimately related to navigational safety at sea. Lack of knowledge on the manoeuvring behaviour of a ship may result in marine casualties. According to a recent investigation, navigational casualties from collision and grounding incidents accounted for more than 40% of all marine incidents. The main underlying cause was examined and found to be inadequate human action (EMSA, 2020). Without a doubt, these accidents stem from inappropriate manoeuvring actions by navigation officers. In addition, a study by Zhang and Xiao-Ming (2017) indicates that a large number of marine accidents are associated with rough sea weather. In general, navigators can access manoeuvring information to better understanding a ship's manoeuvrability, which has been measured by full-scale sea trials or model-scale experiments. However, this manoeuvring information cannot cover a wide range of real sea conditions because the manoeuvring tests have been required to be conducted in the calmest possible weather conditions as recommended by ITTC (2002). Given the limited manoeuvring information available for navigators, more extensive investigations into a ship's manoeuvring in real sea states is necessary for safe manoeuvring.

Ships are exposed to various environmental loads such as waves, wind, and currents during manoeuvring, which affect the manoeuvring performance during operation. Among these external disturbances, waves have been recognised as the most influential factor leading to substantial changes in a ship's manoeuvring performance. Thus, it is of paramount importance to understand the manoeuvring characteristics of a ship in waves to ensure its navigational safety at sea. In response, the ITTC Advisory Council proposed a Specialist Committee which is responsible for studying the manoeuvring characteristics of vessels in waves. The technical committee was established by the 28th International Towing Tank Conference (ITTC, 2017). Following this, the SIMMAN

workshop was organised in 2020 to discuss not only ship manoeuvring in calm water but also ship manoeuvring performance in waves. Despite the growing awareness of the importance of knowledge of a ship's manoeuvrability in waves, such manoeuvring in waves has still not been extensively studied. This may be due to the high cost of the experimental equipment or the limitation of the theoretical method for resolving complicated wave problems.

Traditionally, a physical basin or tank has been used to assess the manoeuvrability of a ship by carrying out Planar Motion Mechanism (PMM) or free-running tests to provide reliable data about ship manoeuvring performance. However, a lot of manoeuvring experiments have been largely confined to the calm water condition. For ship manoeuvring in waves, the demanding requirements of the test facility may include free running equipment, wave basin, and other costly devices. As a result of these impediments, only a few institutes have been able to carry out such tests in waves. For example, the Iowa Institute of Hydraulic Research (IIHR) (Araki et al, 2012; Sanada et al, 2013; Sanada et al, 2019) and Hiroshima University (HR) (Yasukawa et al, 2021) conducted standard manoeuvring tests in waves.

Apart from experiments, theoretical methods have been employed for the prediction of ship manoeuvring characteristics in waves. Over years of research, mathematical models together with potential flow theory have been used for these manoeuvring analyses. Skejic and Faltinsen (2008) adopted a two-time scale model to combine the high-frequency seakeeping motion and the low-frequency manoeuvring motion in regular waves. The ship manoeuvring analysis was based on the nonlinear slender-body theory, while the mean second-order wave loads in incident waves were predicted by the potential flow theory. Seo and Kim (2011) used a 4 degrees of freedom MMG (Mathematical Manoeuvring Group) model along with a time-domain ship motion code to solve the wave-body interaction problem, in which the mean drift forces and moments were considered by using a direct pressure integration method. Their results showed similar trends with the experiments but with some discrepancies. Afterwards, a similar method was adopted by Subramanian and Beck (2015), Zhang and Zou (2016), and Paroka et al., (2017) to estimate the manoeuvring behaviour of a ship in waves. However, all the comparison between the numerical results and experimental data showed considerable discrepancies. It is concluded from these studies that the mathematical model combined with potential flow theory may not be ideal to assess ship manoeuvring characteristics in waves. This is because the simplified mathematical model cannot accurately resolve the complex interactions between hull, rudder, propeller, and waves during manoeuvring. In addition, potential flow theory neglects the influence of viscosity, and thus suffers some limitations due to lack of physics related to viscous effects, as discussed in Mofidi and Carrica (2014).

Meanwhile, as computational capabilities have increased more and more, Computational Fluid Dynamics (CFD) has been used to obtain better predictions through free-running simulations (e.g. turning and zigzag manoeuvres). It is worth mentioning that this method can provide more reliable prediction of ship manoeuvring motions than theoretical methods because it is able to incorporate both viscous and rotational effects in the flow problem. This approach can also accurately estimate the effects of the complex interaction between the hull, propeller, and rudder. Mofidi and Carrica (2014) carried out CFD simulations of zigzag tests of the KCS container model with moving rudder and rotating propeller. The standard 10°/10° zigzag manoeuvre and the modified 15°/1° zigzag manoeuvre in calm water was simulated. In their work, the simulation results agreed well with the experimental results, but the authors highlighted that the computational cost in directly calculating the free running conditions is considerably high for many applications. Broglia et al. (2015) presented CFD simulations of the turning circle manoeuvre of a naval supply vessel characterised by a fully appended single rudder/twin screws configuration. Each propeller was modelled by an actuator disk, well modified to resolve oblique flow effects. The results obtained using the CFD methods were compared to free-running model tests and showed a good agreement in respect of both kinematic and trajectory parameters. Similarly, an implementation of the dynamic overset method in OpenFOAM with application to KCS self-propulsion and zigzag manoeuvres was presented by Shen et al. (2015). Their CFD results for self-propulsion and zigzag tests were found to be in good agreement with the experimental data, although relatively coarse grids were applied in their CFD simulations.

Moreover, the direct CFD approach can be extended to the ship manoeuvring problem in waves. Wang et al. (2017) reported results from the direct CFD simulation of the free running ONR Tumblehome ship under course keeping control in waves. The overset grid method and a 6 degrees of freedom (DOF) module were used to simulate the free running ship motions with twin rotating propellers and moving rudders. To achieve course keeping demand, a course keeping control module was developed to perform the course keeping simulation. The

simulation results were validated against the benchmark data from the Tokyo 2015 CFD Workshop on ship hydrodynamics. Good agreements were obtained for course keeping manoeuvres. Wang et al. (2018) further performed CFD simulations of the standard zigzag manoeuvre in both calm water and waves. The predicted ship motions agreed well with the experimental measurement, which further indicates the feasibility of the CFD approach with dynamic overset methods. For the direct CFD manoeuvring simulations of the KCS at its service speed, the turning circle and zigzag manoeuvres in head waves were conducted using URANS computation by Yuting Jin et al. (2019). However, they did not validate their CFD measurements against any experimental data.

In addition, since the Marine Environment Protection Committee (MEPC) of the International Maritime Organisation (IMO) adopted the interim guidelines for determining minimum propulsion power to maintain the manoeuvrability of ships in adverse conditions (IMO, 2014), the demand for accurately assessing the manoeuvring and seakeeping behaviour of a vessel in adverse weather has significantly increased.

To the best of our knowledge, there exists no specific study investigating the effects of wave directions on the course-keeping ability and the turning manoeuvres of a ship. Therefore, the aim of this study is to fill this gap by performing free-running CFD simulations in waves of different directions. In this study, direct CFD models based on Unsteady Reynolds Averaged Navier-Stokes (URANS) were developed. Each simulation was modelled in five different wave directions, namely head wave, bow wave, beam wave, quartering wave, and following wave. The simulations of self-propulsion in waves were first conducted to achieve the approach speed. Following this, the course-keeping tests were performed to evaluate the course-keeping qualities in different wave headings. Finally, the turning abilities in waves were investigated by conducting the standard turning circle test.

The remainder of this paper is organised as follows. In the next section, we briefly illustrate the methodology of the direct CFD manoeuvre simulation in waves. Section 3 presents a list of simulation cases and the ship particulars applied to the current CFD model. Afterwards, the specific numerical setup of the CFD model is introduced in Section 4. Following this, Section 5 provides details of the simulation results from this work, including validation and verification studies. Finally, a brief summary of the study's findings is provided in Section 6, together with suggestions for future research.

2. Methodology

Fig. 1 illustrates the research methodology followed in this work for the direct CFD manoeuvring simulations. This methodology was organised under four major steps. The first step deals with the fundamental components of the ship model to enable the ship to freely move in 6 degrees of freedom with a moving rudder and an actuator disk. The dynamic overset grid method was applied to the hull and rudder region. The surrounding overset grids of the hull region can handle the large-amplitude motion of a ship in waves, and rudder overset grids are able to control rudder movement towards the port or starboard side for the ship manoeuvre. A body force approach was applied to represent the effect of a rotating propeller. It is noteworthy that the PID feedback controller was utilised to act on the rudder rotational speed to allow the ship to sail straight under the influence of external disturbances. The second step is related to the environmental condition adopted in this paper, particularly regular waves. The manoeuvring computation cases applied to the current CFD model were combined with different incident wave directions: *1) Head sea, 2) Bow sea, 3) Beam sea, 4) Quartering sea, 5) Following sea*. In the third step, the self-propulsion condition is achieved before conducting the standard ship manoeuvring tests. Once the self-propulsion computations were fulfilled, the actuator disk rotational speed was frozen. Finally, the ship manoeuvring module was developed in order to perform the standard ship manoeuvres after reaching the target speed.

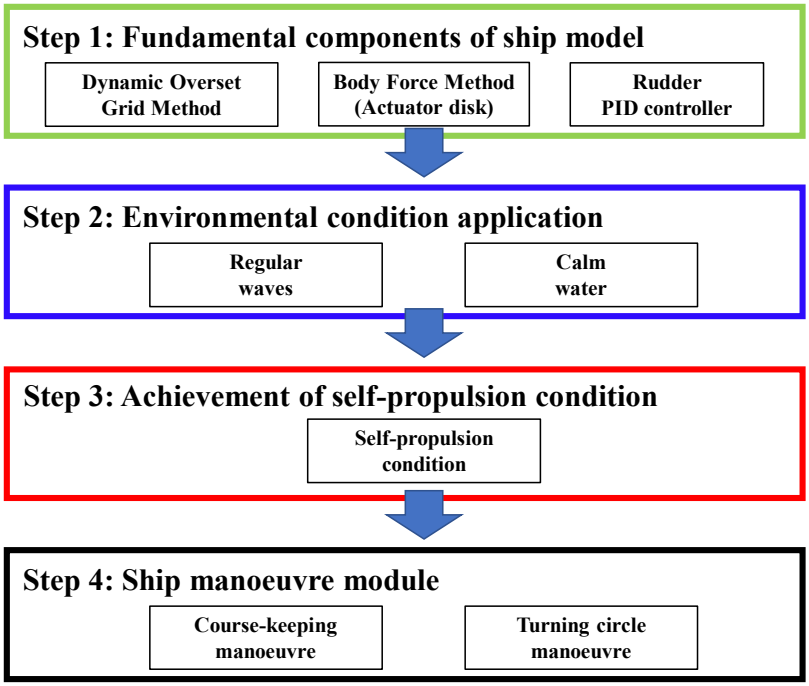


Fig. 1 An overview of research methodology for the direct CFD manoeuvring simulations.

3. Ship geometry and conditions

A KCS container ship model (scale factor of 75.24) appended with a single rudder and an actuator disk was modelled in the CFD simulations. The principal characteristics for the KCS model are listed in Table 1(SIMMAN, 2020).

The ship manoeuvring analysis was carried out using six different conditions, as described in Table 2 and Fig. 2. The approach speed of 14.5 knots (7.459 m/s) in full scale was chosen for the ship manoeuvre in both calm and head sea conditions (case 1 and 2) because of the existence of the available experimental results for comparison while the service speed is 24 knots (12.345 m/s. It should be noted that, despite using the same propeller speed, the approach speeds for the rest of the simulations (case 3-6) vary depending on the incident wave directions, (13.38 RPS).

A wave with height $H = 0.048\,m$ and period $T = 1.4\,s$ in the model scale was employed for the simulations (length ratio $\lambda/L_{BP} = 1.0$ and wave steepness $H/\lambda = 0.016$). These values correspond to a wave height of $3.61\,m$ and period of $12.14\,s$ in full-scale. The encounter frequency of the wave, ω_e, was calculated as $\omega_e = \omega[1 - (\omega U cos\mu)/g]$, where ω is the wave frequency, U is the ship forward speed, μ is the ship's heading angle relative to the wave direction, and g denotes the gravitational acceleration.

Table 1 The main particulars of the KCS model used in this study.

Main particulars	Symbols	Model scale (1:75.24)
Length between the perpendiculars	$L_{BP}(m)$	3.057
Length of waterline	$L_{WL}(m)$	3.0901
Beam at waterline	$B_{WL}(m)$	0.4280
Draft	D(m)	0.1435
Displacement	$\Delta(m^3)$	0.1222
Block coefficient	C_B	0.651
Ship wetted area with redder	S (m^2)	1.6834
Longitudinal centre of buoyance	$\%L_{BP}$, fwd+	-1.48

The metacentric height	GM(m)	0.008
Radius of gyration	K_{xx}/B	0.49
Radius of gyration	$K_{yy}/L_{BP}, K_{zz}/L_{BP}$	0.25
Propeller diameter	$D_P(m)$	0.105
Propeller rotation direction (view from stern)		Right hand side
Rudder turn rate	(deg./s)	20.1

Table 2 The simulation cases to which the CFD model is applied.

Case no. C	Approach speed U_0 (m/s)	Propeller rev. (RPS)	Wave height H (m)	Encounter Angle μ (degrees)	Encounter Freq. ω_e (rad/s)	Encounter Period T_e (s)	Wave/ship length λ/L_{BP}
1	0.86	10.56	Calm water	- (Calm sea)	-	-	-
2	0.86	13.38	0.048	180 (Head sea)	6.257	1.004	1.00
3	0.95	13.38	0.048	225 (Bow sea)	5.863	1.071	1.00
4	1.07	13.38	0.048	270 (Beam sea)	4.490	1.399	1.00
5	1.03	13.38	0.048	315 (Quartering sea)	2.987	2.103	1.00
6	1.05	13.38	0.048	0 (Following sea)	2.332	2.694	1.00

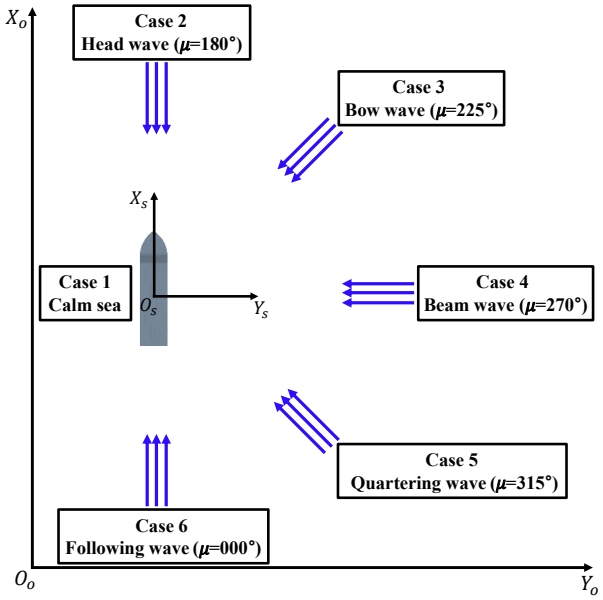


Fig. 2 Schematic view of all the cases applied in this study for ship manoeuvrability.

4. Numerical modelling

The direct manoeuvring simulations of a vessel for this study were performed by using the commercial CFD software STAR-CCM+, version 15.04. This chapter will discuss the specific numerical methods employed in this paper to compute the direct CFD simulations.

4.1. Governing equations

In order to solve the governing equations in this paper, an Unsteady Reynolds-Averaged Navier-Stokes (URANS) method was used. For unsteady incompressible flows, the averaged continuity and momentum equations are expressed in tensor notation and Cartesian coordinates as in the following expressions (Ferziger and Peric, 2020):

$$\frac{\partial(\rho\bar{u}_i)}{\partial x_i} = 0 \qquad (1)$$

$$\frac{\partial(\rho\bar{u}_i)}{\partial t}+\frac{\partial}{\partial x_i}\left(\rho\bar{u}_i\bar{u}_j+\rho\overline{u'_\iota u'_j}\right)=-\frac{\partial\bar{p}}{\partial x_i}+\frac{\partial\bar{\tau}_{ij}}{\partial x_i} \qquad (2)$$

where ρ is the fluid density, $\bar{u}_i$ is the averaged velocity vector, x_i (i=1, 2, 3) are the Cartesian coordinates, $\rho\overline{u'_\iota u'_j}$ is the Reynolds stresses, $\bar{p}$ is the mean pressure and $\bar{\tau}_{ij}$ are the mean viscous stress tensor components. This stress tensor for a Newtonian fluid can be given in Eq. (3)

$$\bar{\tau}_{ij}=\mu(\frac{\partial\bar{u}_i}{\partial x_j}+\frac{\partial\bar{u}_j}{\partial x_i}) \qquad (3)$$

in which μ is the dynamic viscosity. The approach of RANS for turbulent flows requires that the Reynolds stresses in Eq (4) be appropriately modelled. A common method uses the Boussinesq theory:

$$-\rho\bar{u}_i\bar{u}_j=u_t\left(\frac{\partial\bar{u}_i}{\partial x_j}+\frac{\partial\bar{u}_j}{\partial x_i}\right)-\frac{2}{3}(\rho k+u_t\frac{\partial\bar{u}_k}{\partial x_k})\delta_{ij} \qquad (4)$$

where u_t is the turbulent eddy viscosity, and k and δ_{ij} are the turbulence kinetic energy and Kronecker delta, respectively.

To compute the fluid flow problem numerically, the solution domains were subdivided into a finite number of adjacent control volumes using a finite volume (FV) method. The governing equations were converted to algebraic equations that can be solved numerically according to a control-volume-based technique. The FV method is appropriate for complex geometries. The suitability of the FV method is due to its capability of accommodating any type of grid. The URANS solver employs a pressure-velocity coupling algorithm where a predictor-corrector approach is applied to connect the continuity and momentum equations.

4.2. Physics modelling

The turbulence model applied in this paper was the Shear Stress Transport (SST) turbulence model (Menter et al., 1994), which effectively blends a $k-\varepsilon$ model in the far-field with a $k-\omega$ model near the wall. Menter's SST model has also been applied in several studies performed in the same area (Mofidi and Carrica, 2014; Wang et al., 2018; Liu et al., 2020). An all-y+ wall treatment was selected, achieving a final y+ value of higher than 30 along the hull for an accurate prediction of the flow across the wall boundary.

The "Volume of Fluid" (VoF) multiphase model was employed to represent the free surface with either a flat or regular wave condition. It also provides field functions that can be used to initialise the VoF computation and to describe proper profiles at boundaries. Tezdogan et al. (2015) outlined that the VoF method offers high numerical efficiency and is appropriate for resolving flows in which each phase forms a large structure, with a low overall contact area between the different phases. This model has been typically used along with the 6 DOF motion model for ship hydrodynamics applications. The grid refinement near the free surface area was carried out to capture accurately the variations of air-water interfaces. It is worth noting that a second-order convection scheme was applied in all CFD simulations to compute the air-water interfaces with a sustainable level of accuracy. It should also be illustrated that in the RANS approach, a segregated solution method, which solves each discrete governing equation implicitly, was chosen throughout all simulations. A SIMPLE-type-algorithm was applied for the overall numerical calculations in this paper.

To simulate the standard manoeuvres of the KCS, a Dynamic Fluid Body Interaction (DFBI) module integrated in the 6 DOF motion solver was utilised with the ship free to move in 6 degrees of freedom. This method computes the resultant force and moment acting on the ship and calculates the governing equations of motion to determine the new position of the ship. The governing equations of motion can be described as follows:

$$\frac{\mathrm{d}(m\mathbf{V})}{dt}=\mathbf{f} \qquad (5)$$

$$\mathrm{M_b}\frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{dt}}+\boldsymbol{\omega}\times\mathrm{M_b}\,\boldsymbol{\omega}=\mathbf{m} \qquad (6)$$

where m indicates the mass of body, $\mathbf{V}$ is the velocity vector of its centre of mass, $\mathbf{f}$ is the resultant force vector acting on the ship, $\mathrm{M_b}$ stands for the tensor of moments of inertia in the body-fixed coordinate system, $\boldsymbol{\omega}$ is the angular velocity vector of the body and $\mathbf{m}$ denotes the resultant moment vector acting on the body.

The body force propeller method was applied to represent the features of the propeller through body forces in the RANS computation. The axial and tangential forces of the modelled propeller and its effects on the flow are calculated using this model. The integration of these forces over the disk gives the thrust and torque of the propeller, which can then be combined with the DFBI module. The propeller effects are treated as body force terms in the governing equations. This simplified disk model can reduce the computational resources to perform the simulations. In order to conduct the modelling of an actuator disk, numerous input parameters are required including propeller dimensions, the rotational speed, and performance curve results. In this paper, input variables were determined according to the propeller experimental results carried out by Hiroshima University. Throughout all simulations, the propeller-induced velocity was corrected to model the effects of the propeller efficiently based on the induced velocity correction property described in Neitzel et al. (2015).

5th-order Stokes waves were applied for the wave simulations. The theory of the 5th-order wave is based on the study by Fenton (1985). According to CD-Adapco (2020), this wave more closely resembles a real wave than a wave that is generated by the first order method. The first order wave mentioned here is modelled with a first order approximation to the Stokes theory. This approximation generates waves that have a regular periodic sinusoidal profile.

4.2.1. Coordinate systems

Four coordinate systems were adopted to perform the direct manoeuvring simulations of a vessel in 6 DOF as follows: (1) Earth-fixed system ($O_O - X_O Y_O Z_O$), (2) Ship-fixed system ($O_S - X_S Y_S Z_S$), (3) Propeller-fixed system ($O_p - X_p Y_p Z_p$), (4) Rudder-fixed system ($O_r - X_r Y_r Z_r$). The Earth-fixed coordinate system is an inertial coordinate frame, which was fixed to a specific point with both translation and rotation motion not allowed. The ship-fixed system was located at the centre of mass of a vessel and the 6 DOF motion equations were solved under its coordinate system. A propeller-fixed system was introduced for the positioning of the actuator disk. A rudder-fixed coordinate was applied to control the rudder angle for the ship manoeuvring. It is of note that the rudder stock refers to the axis about which the rudder of a ship rotates. In this work, the Z_r axis indicates the rudder stock axis. Therefore, the rudder blade rotates around the Z_r axis. All coordinate systems used in this paper are shown in Fig. 3.

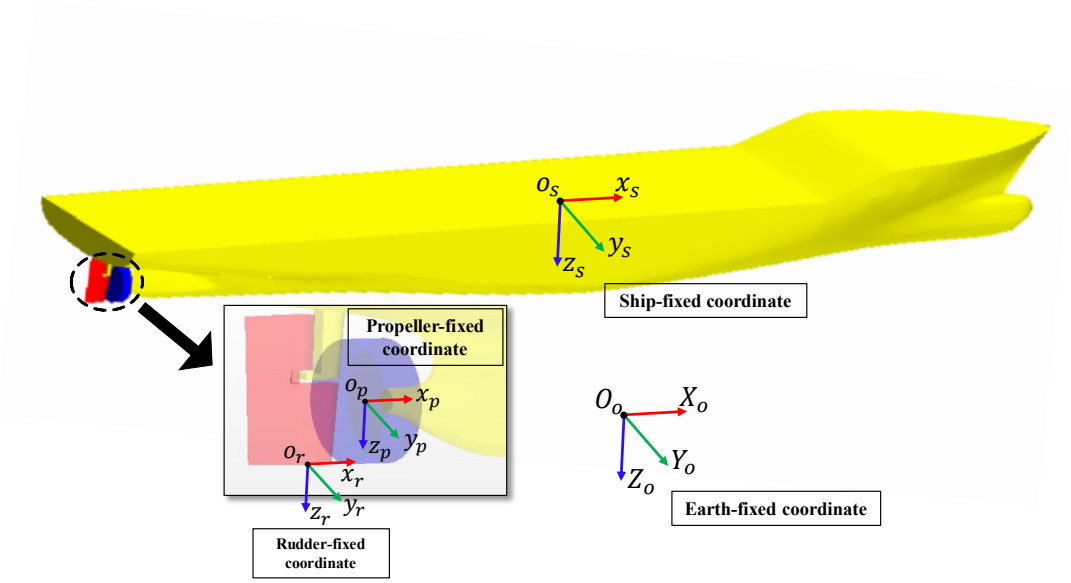
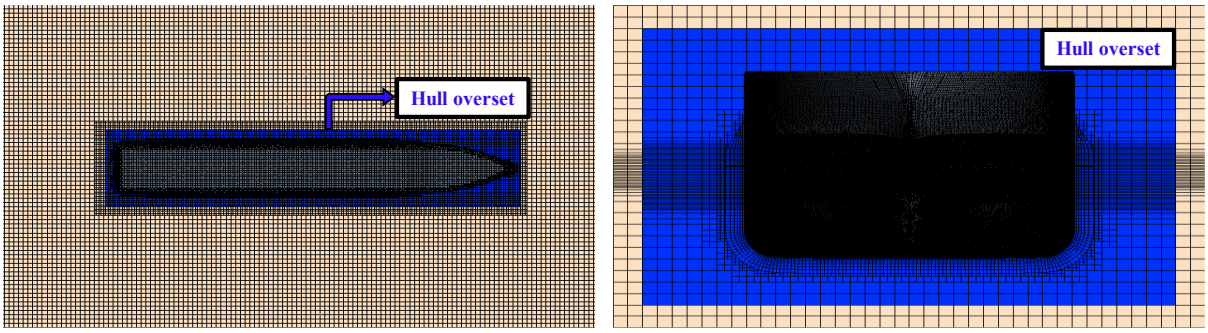


Fig. 3 The coordinate systems of the ship manoeuvring simulation used in this study.

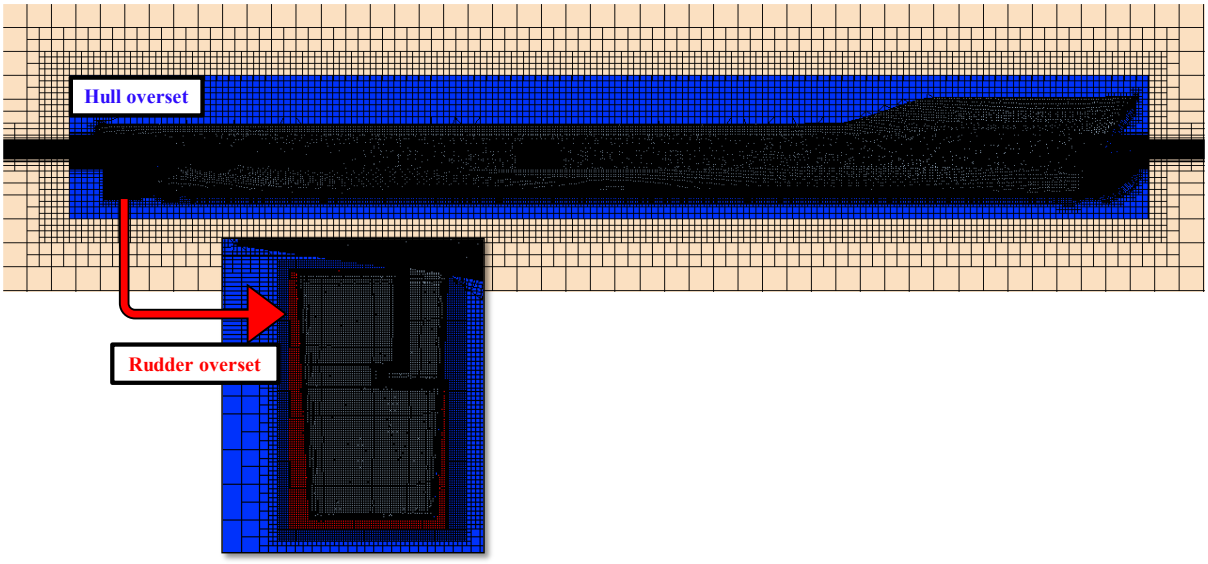
4.2.2. Mesh generation

The computational domain meshes were generated by applying the automatic meshing facility in STAR-CCM+, which automatically generates the surface or volume grids suitable for the finite volume method. A surface remesher tool was used to enhance the overall quality of part surfaces and optimise the mesh quality for the volume grids. A trimmed cell mesher was used for the volume meshes. The resulting mesh was made up mainly of hexahedral cells with trimmed cells neighbouring the surface. The prism layer mesh model was adopted to provide orthogonal prismatic cells adjacent to the wall surfaces or boundaries, which is required to resolve near wall flow accurately. In addition, additional meshing refinements were carried out in order to obtain meaningful and reliable CFD results. The refinements were conducted in the critical regions such as the area immediately surrounding the ship hull, the tight gap parts between the rudder blade and the horn, and the free surface.

To simulate ship motions in waves, the free surface mesh refinement was carried out by generating 80 cells per wavelength and 20 cells per wave height. This is based on the guidelines for ship CFD applications from ITTC (2011). According to these recommendations, a minimum of 80 cells wavelength was required on the free surface, while a minimum of 20 cells should be used in the vertical direction where the free surface is expected. Fig. 4 shows cross-sections of the meshed domain whereas Fig. 5 demonstrates the surface grids and prism layers on the KCS hull and the rudder for the manoeuvre simulations. It is of note that the grid refinement was applied to the whole free surface area (x, y, and z directions, respectively) to ensure a reliable solution for such a complex wave problem. The total grid number applied to the calm water (case 1) and wave (case 2-6) simulations is 6.90 million and 7.19 million, respectively.

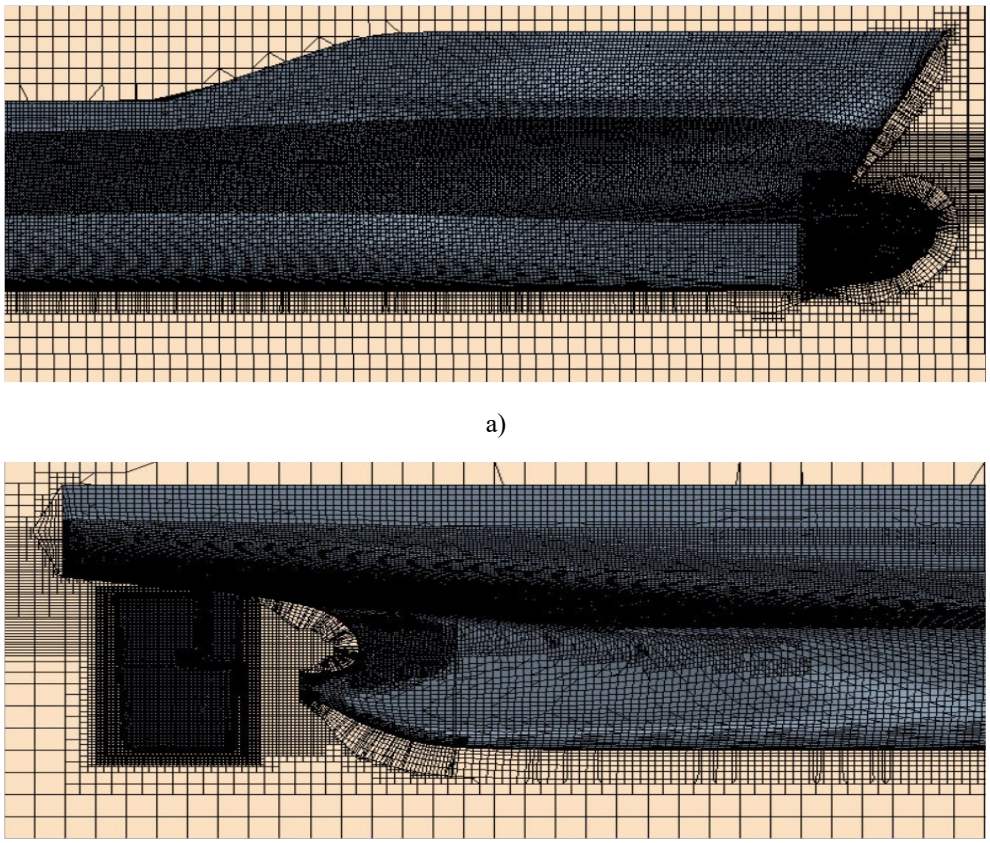


a)



b)

Fig. 4 Computational mesh: a) Top and front view b) profile view of the domain



a)

b)

Fig. 5 Surface grids and prism layers on the a) bow b) stern of the KCS hull and rudder.

In order to perform the CFD simulations of a free running ship, the dynamic overset grid method was adopted in this study. Wang et al. (2017) pointed out that the dynamic overset grid is the essential point for simulations of complex ship motions with moving components, such as rudders and propellers. With this method, the arbitrary ship motion in 6 DOF could be accounted for while controlling the single rudder in order to carry out ship manoeuvres. In this work, the solution domain was divided into three regions: background grids, overset grids around the ship hull, and overlapping grids surrounding the moving rudder. With the overset method, it should be stated that the domain connectivity process must be carried out based on the hole-cutting approach. The hole-cutting operation is the coupling of the overset region with the background region through an overset interface by which the interpolation of dependent variables can be implemented between donor and acceptor cells. The overset interfaces were created using the Overset Hierarchy property, which controls the region priority for the hole-cutting process. A general view of the overset mesh configuration with the KCS hull and rudder model is presented in Fig. 6.

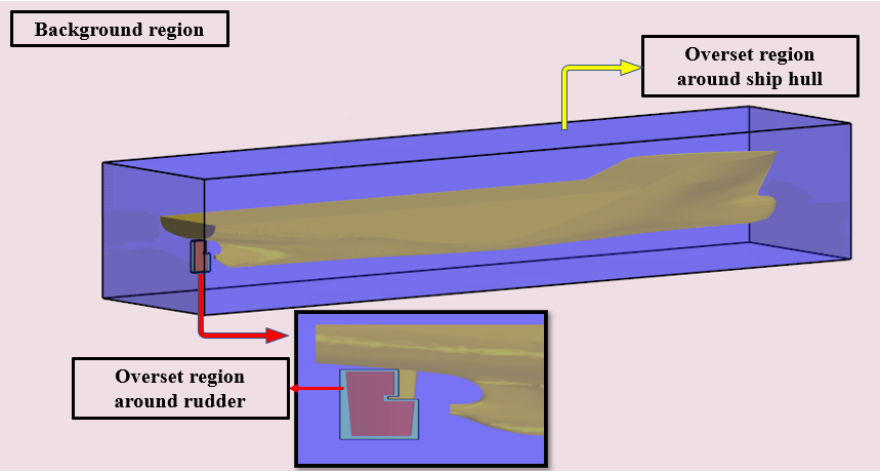


Fig. 6 Overset mesh configuration applied in this work's CFD simulations.

4.2.3. Determination of the time step

There are several approaches to choose the time step of a CFD simulation for stability in the numerical solution. The Courant number (CFL) condition is the most widely used to evaluate the time step requirements of a transient simulation for a given grid size and flow velocity. The CFL number formula is defined as below:

$$\mathrm{CFL} = \frac{U\Delta t}{\Delta x} \quad (7)$$

where U denotes the flow velocity, Δt indicates a representative time step of the simulation and Δx is the mesh cell dimension. It is generally equal to 1 or less for numerical stability.

In addition to the CFL condition, the time step can be decided by the flow features. It is of note that the ITTC (2011) recommends that at least 100 time steps per period for regular waves should be used for the wave simulation. In this paper, therefore, the time step size of all the simulations for the ship manoeuvre was set to $\Delta t = 0.005$, which is two times lower than the recommended guidelines of ITTC (2011) to ensure a reliable solution.

4.2.4. Computational domain and boundary conditions

For accurate resolution of flow around structures, it is imperative in CFD calculations to apply the proper initial and boundary conditions that provide a good approximation to the solution of the problem (Date and Turnock, 1999). Two types of computational domains were created for this work's ship manoeuvre simulations: one for calm water and the other for waves. On each computational domain, different initial and boundary conditions were chosen to represent the KCS model being manoeuvred in the calm water or waves conditions.

A schematic illustration of the computational domain with the notations showing applied boundary conditions is presented in Fig. 7. The boundary conditions for moving surfaces were set as non-slip walls. The top of the computational domain was modelled as a pressure outlet condition. The velocity inlet boundary condition was imposed at the upstream (positive x-direction), the downstream (negative x-direction), side boundaries (the two opposite faces at the y-direction of the domain), and bottom walls to avoid non-physical effects. For the calm water simulation, the damping capability of the software was applied at the up-and-down stream boundaries and side walls to prevent undesired Kelvin wave reflection. For the wave simulations, on the other hand, the forcing method was applied on these four boundaries. As mentioned by Ferziger and Peric (2020), the velocity and the volume fraction of water are forced towards the theoretical solution of Fenton (1985) in the forcing zone. Thus, the ship-induced disturbance of incoming waves gradually vanishes in the forcing zone, which helps to avoid reflections of waves from the boundaries. The forcing zone applied throughout all the simulations in waves are depicted in Fig. 8, in which relative zone distance indicates the relative distance between a cell and the nearest boundary of the forcing zone (CD-Adapco, 2020).

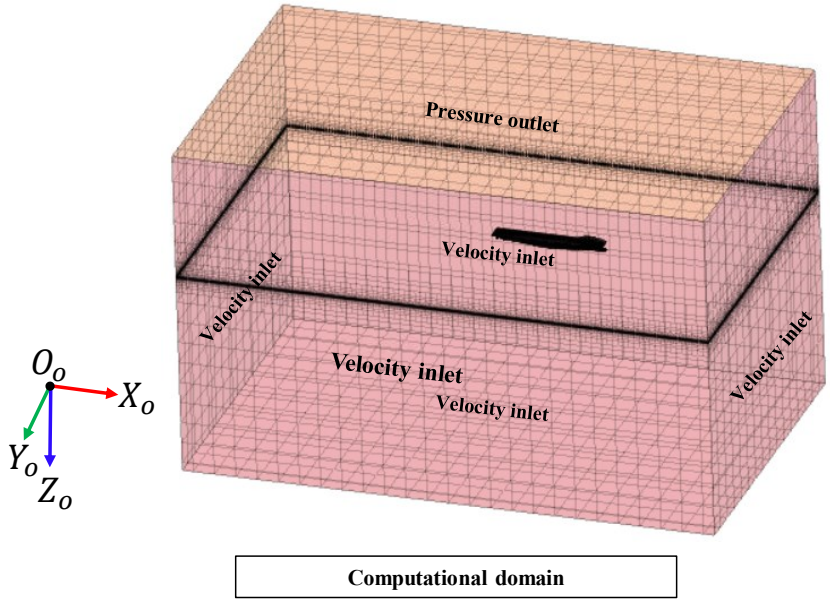


Fig. 7 Schematic views of the computational domain applied for the calm water and wave simulations.

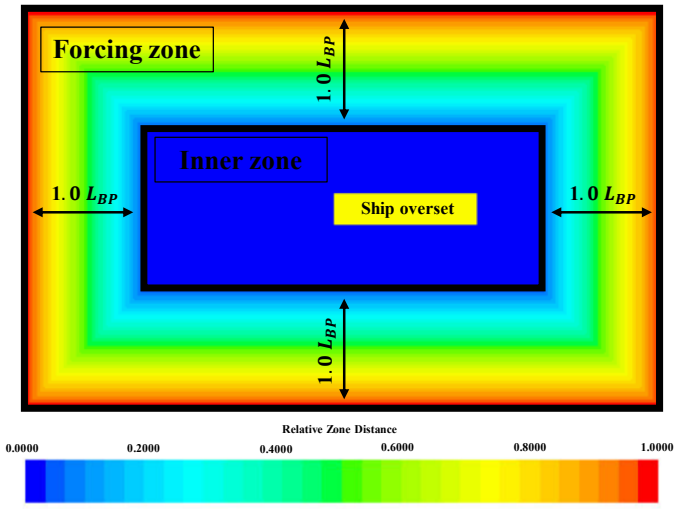


Fig. 8 A schematic view of the forcing zone in the CFD simulations in waves

ITTC (2011) emphasises that the computational domain boundaries should be positioned at a sufficient distance away from the ship to minimise their influence on the solution. The locations of the boundaries were determined based on previous research performing wave simulations. Table 3 shows the findings of all boundary locations. The dimensions of the domain used throughout the direct manoeuvring simulations are shown in Fig. 9.

Table 3 Boundary distances from the ship in similar previous work.

References	Directions				
	Upstream	Downstream	Up	Bottom	Transverse
Tezdogan et al. (2015)	$1.15L_{BP}$	$4.5L_{BP}$	$1L_{BP}$	$2.3L_{BP}$	$2.5L_{BP}$
Wang et al. (2017)	$1.5L_{BP}$	$3.0L_{BP}$	$0.5L_{BP}$	$1.0L_{BP}$	$1.5L_{BP}$
The current CFD	$1.5L_{BP}$	$2.5L_{BP}$	$1.1L_{BP}$	$1.9L_{BP}$	$1.5L_{BP}$

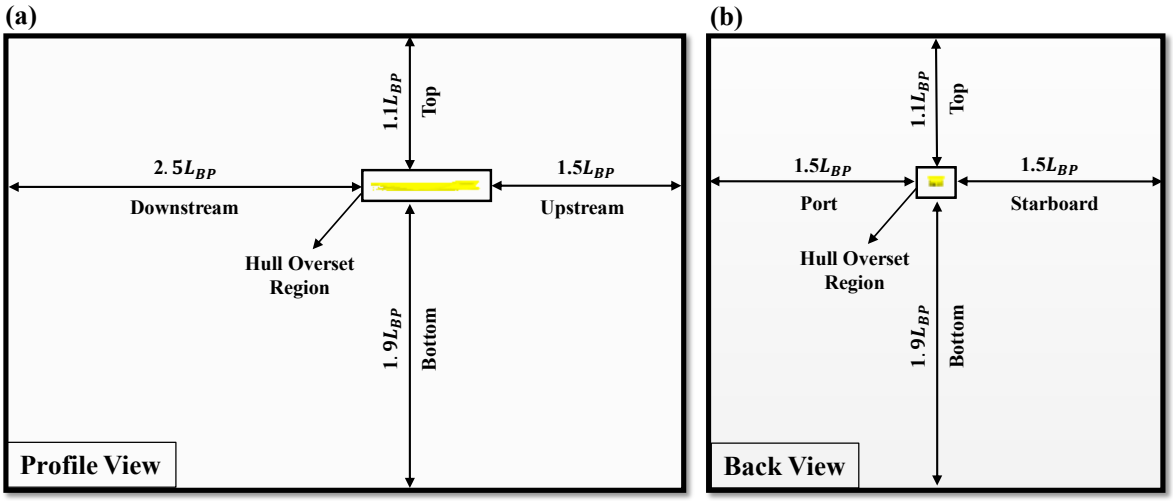


Fig. 9 The dimensions of the computational domain for ship manoeuvring simulations, (a) profile view of the domain, (b) back view of the domain.

The background domain was forced to move with 3 DOF (surge, sway, and yaw) in order to allow a relatively static motion between the ship overset and background regions, which is depicted in Fig. 10. It should be kept in mind that the velocity components of the 3 DOF background domain were set to the velocities of the ship surge, sway and yaw motions. The key advantage of this is that required cell numbers in the background region can be significantly decreased for ship manoeuvre computations and thus reduce the computational demand. Otherwise, the background domain should be enlarged considerably in order to encompass the expected travel path of the ship during ship manoeuvre simulations.

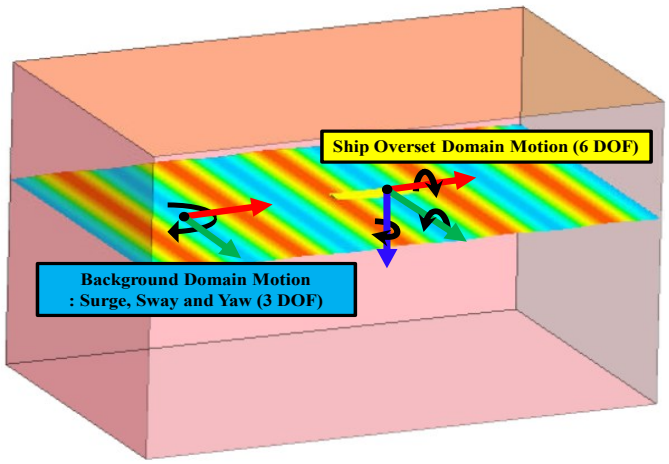


Fig. 10 The view of the background domain and ship overset domain motion.

4.2.5. Course keeping control and turning manoeuvring module

Throughout all the simulations in this study, numerical computations were conducted for a standard turning circle manoeuvre with 35º rudder angle moving to starboard. For measuring ship manoeuvrability, ITTC (2002) recommends that the ship should move forward at a constant speed (thus a constant propeller speed) and be with a minimum yaw rate (steady course) prior to the test execution.

Hence, it is of great importance to keep the straight course of the ship with a constant propeller speed before

starting rudder execution for the standard ship manoeuvres such as a turning circle test. For this reason, a feedback controller was applied to regulate the rudder angle and consequently the ship's yaw angle. The feedback controller enables the ship to sail straight in the presence of external forces generated by waves, which can minimise the deviation of a ship's course. Unfortunately, in this study, there are no available experimental data for the rudder controller from Hiroshima University, and thus the widely used Proportional-Integral-Derivative (PID) controller in industrial control systems was applied. The rudder angle is controlled by the PID feedback controller, where the rudder is moved to port or starboard. The control function is given by:

$$\delta(t) = K_p e(t) + K_i \int_0^t e(t)dt + K_d \frac{de(t)}{dt} \quad (8)$$

$$e(t) = \psi(t) - \psi_c \quad (9)$$

where $\delta(t)$ is the rudder angle, $\psi(t)$ is the instantaneous yaw angle at a given time, ψ_c is the target yaw angle which was set to 0 to make the ship advance straight. K_p, K_i, and K_d indicate the proportional, integral, and derivative control gains, respectively. It is crucial to find appropriate values of the control gains to have good course keeping ability since it is not easy to determine the optimal values. In this paper, these values were chosen as $K_p = 5, K_i = 0.05, and\ K_d = 3$ after several trials and errors and the same values were applied for all manoeuvre simulations to compare the course keeping ability in different waves.

For a standard turning circle manoeuvre, a starboard 35º turning test module was introduced throughout all the simulations. The rudder angle $\delta(t)$ is controlled by the following expressions:

$$\delta(t) = \begin{cases} \max(0, kt), \delta \leq 35 \\ 35 \end{cases} \quad (10)$$

where k is the maximum rudder rate. For the KCS model in this study, the maximum rudder rate is $k = 20.1°/s$ corresponding to $2.32°/s$ in full scale. As mentioned earlier, the rudder angle is turned to 35 degrees and kept at this angle until the ship completes a turning circle test.

5. Results and discussion

The following section aims to provide details of the simulation results obtained during this study, and will compare the results with the available experimental data. Then, it will provide a detailed discussion on the examination of the numerical results. This section is composed of six main sub-sections, and each of them will explain our research findings in detail from different aspects. The time to complete the computation was approximately 336 wall clock hours with 40 CPUs for each manoeuvring simulation.

5.1. Open-water characteristic

Before conducting the manoeuvring simulations, the open-water test of the modelled actuator disk was simulated and compared with the experiments performed by Hiroshima University. Table 4 and Fig. 11 show the CFD calculation results and comparison results, where J represents the advance ratio, K_T is the thrust coefficient, K_Q is the torque coefficient, and η_0 is the efficiency of the propeller. As can be seen from the comparison, the actuator disk based on the body force method predicted the performance curves reasonably well and the obtained results agreed well with the experiments. These results revealed that the actuator disk is capable of estimating the open-water curves in terms of the thrust and torque coefficient. This actuator disk model was used for all the manoeuvring simulations.

Table 4. Propeller open water test results.

J	CFD			Experiment (SIMMAN, 2020)			Error of K_T (%)	Error of K_Q (%)	Error of η_0 (%)
	K_T	K_Q	η_0	K_T	K_Q	η_0			
0.0455	0.4442	0.0655	0.0491	0.4659	0.0685	0.0492	4.66	4.41	0.26

0.1433	0.4107	0.0611	0.1532	0.4398	0.0648	0.1546	6.61	5.71	0.97
0.2203	0.3900	0.0585	0.2337	0.3993	0.0597	0.2345	2.32	2.01	0.33
0.3085	0.3492	0.0533	0.3216	0.3551	0.0540	0.3225	1.66	1.41	0.26
0.3926	0.3075	0.0481	0.3988	0.3083	0.0482	0.3990	0.26	0.23	0.03
0.4884	0.2626	0.0424	0.4808	0.2605	0.0421	0.4800	-0.81	-0.64	-0.16
0.5948	0.2042	0.0351	0.5502	0.2010	0.0347	0.5478	-1.59	-1.15	-0.43
0.6849	0.1567	0.0290	0.5890	0.1531	0.0285	0.5843	-2.35	-1.54	-0.79
0.7753	0.1064	0.0218	0.6003	0.1029	0.0213	0.5941	-3.40	-2.34	-1.04
0.8716	0.0501	0.0140	0.4945	0.0480	0.0137	0.4835	-4.43	-2.11	-2.28
0.9593	-0.0037	0.0066	-0.0855	-0.0035	0.0067	-0.0803	-5.71	0.75	-6.52

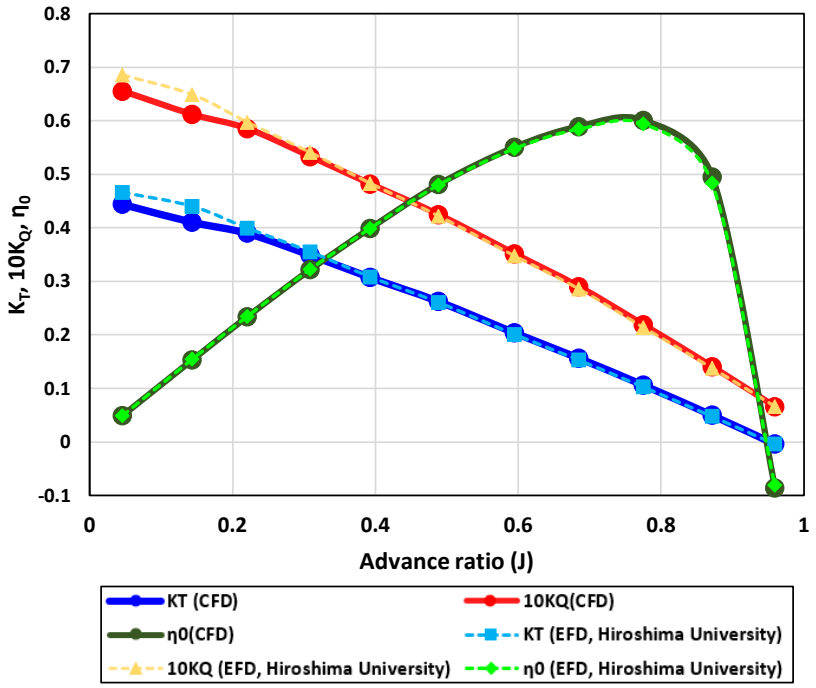


Fig. 11 Propeller open water test results and comparison.

5.2. Verification study

A verification study was carried out to quantify numerical uncertainties of the CFD models. Spatial and temporal convergence analyses were undertaken independently to investigate proper grid-spacing and time step. For mesh convergence behaviour, the Grid Convergent Index (GCI) approach based on the extrapolation of Richardson (1910) was adopted according to Celik et al. (2008) and a brief description is given below. In the same manner, the GCI method can also be employed for a time step convergence study as stated in Song et al. (2020).

Let N_1, N_2 and N_3 denote the total number of grids applied for the numerical computations and h_1, h_2 and h_3 are representative grid sizes. ΔV_i represents the volume of the ith cell and refinement factors r_{21} and r_{32} are defined by the following expressions:

$$r_{21} = \frac{h_2}{h_1} \qquad (11)$$

$$r_{32} = \frac{h_3}{h_2} \qquad (12)$$

$$h_j = \left[\frac{1}{N_j}\sum_{i=1}^{N_j}(\Delta V_i)\right]^{1/3}, \qquad j = 1,2,3 \qquad (13)$$

For a temporal convergence study, the definition of time refinement rate can be given by $r_{21} = \Delta t_2/\Delta t_1$ and $r_{32} = \Delta t_3/\Delta t_2$. It should be taken into account that the refinement factor should be greater than 1.3 according to Celik et al. (2008).

The apparent order of the method, p, is calculated by the formula

$$p = \frac{1}{\ln(r_{21})}\left|\ln\left|\frac{\varepsilon_{32}}{\varepsilon_{21}}\right| + q(p)\right| \qquad (14)$$

$$q(p) = \ln\left(\frac{r_{21}^p - s}{r_{32}^p - s}\right) \qquad (15)$$

$$s = 1 \cdot sgn\left(\frac{\varepsilon_{32}}{\varepsilon_{21}}\right) \qquad (16)$$

In Eq. (16) $\varepsilon_{32} = \phi_3 - \phi_2$ and $\varepsilon_{21} = \phi_2 - \phi_1$ are the differences between coarse-medium and medium-fine solutions, where ϕ_1, ϕ_2, ϕ_3 correspond to the solutions with fine, medium, and coarse input parameters (i.e. grid-size for the mesh convergence study or time-step for the temporal convergence study).

The extrapolated values are defined as:

$$\phi_{ext}^{21} = \frac{r_{21}^p \phi_1 - \phi_2}{r_{21}^p - 1} \qquad (17)$$

In the same way, ϕ_{ext}^{32} is calculated by $\phi_{ext}^{32} = (r_{32}^p \phi_2 - \phi_3)/(r_{32}^p - 1)$.

The definition of approximate relative error (e_a^{21}) and extrapolated relative error (e_{ext}^{21}), respectively, can be described by:

$$e_a^{21} = \left|\frac{\phi_1 - \phi_2}{\phi_1}\right| \qquad (18)$$

$$e_{ext}^{21} = \left|\frac{\phi_{ext}^{12} - \phi_1}{\phi_{ext}^{12}}\right| \qquad (19)$$

Eventually, the fine-grid convergence index is obtained as follows:

$$GCI_{fine}^{21} = \frac{1.25 e_a^{21}}{r_{21}^p - 1} \qquad (20)$$

In this paper, the verification parameters of the advance, transfer, and tactical diameter for the spatial and temporal convergence studies were investigated. These parameters were identified by ITTC (2002) to which reference can be made for more detailed information on the definition of a turning manoeuvre. The relevant indices are illustrated in Fig. 12.

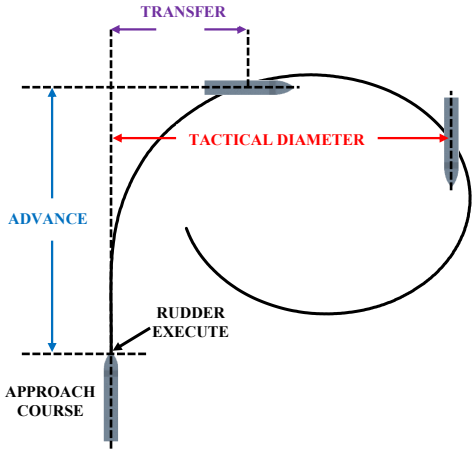


Fig. 12 The definitions of a turning circle.

5.2.1. Spatial convergence study

The head wave simulation (case 2) was chosen for the spatial convergence study. The grid convergence test was performed with three different resolutions of grids, which are regarded as fine, medium, and coarse meshes corresponding to grid numbers of 7,189,514 cells, 4,825,180 cells, and 2,735,883 cells. Table 5 shows the required parameters for the calculation of the spatial discretization error. The ship advance, transfer, and tactical diameter were used as the key variables. As can be seen from Table 5, the numerical uncertainties of 0.11%, 0.28%, and 0.06% were calculated for the computed ship advance, transfer, and tactical diameter, respectively.

Table 5 Calculation of the discretization error for the spatial convergence study, key variables: Advance, transfer, and tactical diameter.

	Advance	Transfer	Tactical diameter
N_1	7,189,514	7,189,514	7,189,514
N_2	4,825,180	4,825,180	4,825,180
N_3	3,153,041	3,153,041	3,153,041
r_{21}	1.14	1.14	1.14
r_{23}	1.15	1.15	1.15
ϕ_1	7.866m	3.089m	7.968m
ϕ_2	7.924m	3.141m	8.024m
ϕ_3	8.598m	3.662m	9.131m
ε_{32}	6.74E 01	5.21E 01	1.11E + 00
ε_{21}	5.82E 02	5.18E 02	5.60E 02
s	1	1	1
e_a^{21}	0.74%	1.68%	0.70%
q	−1.69E 01	−1.61E 01	−1.98E 01
p_a	1.72E + 01	1.62E + 01	2.10E + 01
ϕ_{ext}^{21}	7.86E + 00	3.08E + 00	7.96E + 00
e_{ext}^{21}	0.08%	0.22%	0.05%
GCI_{fine}^{21}	0.11%	0.28%	0.06%

5.2.2. Temporal convergence study

The head wave simulation (case 2) was also selected for the temporal convergence study. Three different time steps were employed based on a uniform refinement ratio of 2, starting from $\Delta t = 0.005s$. Table 6 describes the required parameters of the calculation of the temporal discretization error. The ship advance, transfer, and tactical diameter were used as the key variables. As can be seen from Table 6, the numerical uncertainties of 0.161%, 0.196%, and 0.043% were calculated for the computed ship advance, transfer, and tactical diameter, respectively.

Table 6 Calculation of the discretization error for the temporal convergence study, key variables: Advance, Transfer, and Tactical diameter.

	Advance	Transfer	Tactical diameter
Δt_1	$0.005s$	$0.005s$	$0.005s$
Δt_2	$0.01s$	$0.01s$	$0.01s$
Δt_3	$0.02s$	$0.02s$	$0.02s$
r_{21}, r_{32}	2	2	2
ϕ_1	$7.86582m$	$3.089m$	7.968m
ϕ_2	$7.84181m$	$3.109m$	$7.996m$
ϕ_3	$7.76073m$	$3.214m$	8.313m
ε_{32}	$-8.11\text{E }02$	$1.04\text{E }01$	$3.17\text{E }01$
ε_{21}	$-2.40\text{E }02$	$2.02\text{E }02$	$2.81\text{E }02$
e_a^{21}	0.31%	0.65%	0.35%
p_a	$1.76\text{E} + 00$	$2.37\text{E} + 00$	$3.49\text{E} + 00$
ϕ_{ext}^{21}	$7.876\text{E} + 00$	$3.084\text{E} + 00$	$7.965\text{E} + 00$
e_{ext}^{21}	0.128%	0.157%	0.034%
$GCI_{\Delta t_1}^{21}$	0.161%	0.196%	0.043%

5.3. Validation study

In validation, experimental and numerical results were compared to determine the relative error of the numerical results. It is of note that the fine mesh and fine time step were used for each case. The ship turning manoeuvre in calm sea (case 1) and head sea (case 2) were carried out first for the validation study. The procedure to conduct a turning manoeuvre requires achieving the self-propulsion condition at the approach speed. The target speed was 0.86 m/s (14.5 knots at full scale) in both case 1 and case 2 and the actuator disk speed was frozen after the ship achieved the self-propulsion condition. Following this, turning manoeuvre tests were conducted by controlling the rudder angle according to the turning manoeuvre module. The computed essential manoeuvring indices and hydrodynamic characteristics for the KCS were validated against the available experimental data from Hiroshima University (Yasukawa et al, 2021). It is of note that the experimental uncertainties in the critical manoeuvring quantities are not available in the literature, and thus could not be included in this paper. Therefore, the obtained CFD results were compared with the uncorrected experimental data, assuming that the results from the experiments are sufficiently reliable for the validation study.

The predicted major parameters of the ship turning manoeuvre were compared to those from their experiments (Yasukawa et al, 2021) as shown in Table 7 andTable 8. It is worth noting that the time was shifted to match the beginning of the rudder execution time for the correct comparison with the experimental results. The simulated results were found to be in good agreement with the experiments with errors up to 5% in calm sea and 8% in head sea, respectively. The actuator disk model applied to the current CFD approach was able to accurately represent the effects of the real propeller on the self-propulsion condition with respect to the rotational speed and thrust.

Not only that, but the predicted turning indices, i.e., advance, transfer, tactical diameter as well as time to 90°/180° yaw angle change, also showed reasonable results, with the measured results with errors varying from7.78% to 4.86%. This indicates that the CFD method can resolve the hull-propeller-rudder interaction affecting the ship manoeuvre characteristics with reasonable accuracy during a ship turning manoeuvre.

Table 7 Comparison of the main parameters of a standard turning circle manoeuvre in calm water.

Parameters	CFD	EFD (Yasukawa et al, 2021)	Error (%)
RPS at self-propulsion point	10.56	10.40	1.58
Advance (m)	9.30	9.29	0.01
Transfer (m)	3.95	4.16	4.86
Time for yaw 90 degrees (s)	15.45	15.64	-1.22
Tactical diameter (m)	9.94	9.66	2.89
Time for yaw 180 degrees (s)	31.50	30.50	3.28

Table 8 Comparison of the main parameters of a standard turning circle manoeuvre in head wave.

Parameters	CFD	EFD (Yasukawa et al, 2021)	Error (%)
RPS at self-propulsion point	13.38	13.2	1.36
Advance (m)	7.86	8.20	-4.11
Transfer (m)	3.09	3.35	-7.78
Time for yaw 90 degrees (s)	12.52	13.29	-5.81
Tactical diameter (m)	7.97	8.37	-4.83
Time for yaw 180 degrees (s)	23.74	25.32	-6.24

Fig. 13 and Fig. 14 illustrate the comparisons of the time histories of the ship speeds, ship motions, propeller thrust, and torque during a turning manoeuvre in calm water and regular head waves. It is evident that the simulation results were consistent with the experiments regarding surge velocity and yaw velocity for both calm and head seas, though slight discrepancies were observed for sway velocity in both cases. It is of note that ship surge velocity reduced considerably due to the increased resistance after the rudder execution to starboard $35°$ for the turning manoeuvre, and reached the steady speed of approximately 50% and 60% of the approach speed (0.86 m/s) in calm water and head wave, respectively.

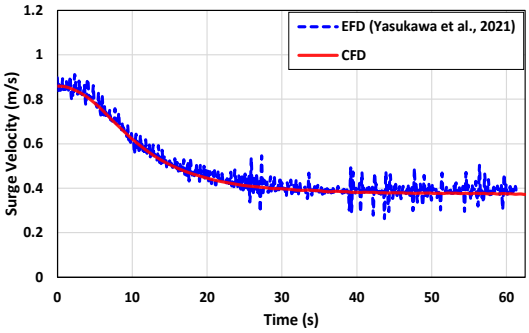
As for the ship seakeeping behaviour in Fig. 13 and Fig. 14 (d - f), the large discrepancy of roll motion between the CFD prediction and experimental measurement was noted, while the pitch and heave motions showed a reasonable match with the experiments. The discrepancy concerning the roll motion may be caused by two possible reasons. The first is the experimental uncertainty of KG (the vertical distance between the keel and the centre of gravity). The uncertain KG position in CFD calculations may give rise to the disagreement between the simulation and experiment since the properties of the roll motion are mainly determined by the height of centre of mass. Another reason is the body force approach applied to the CFD model, which may not be able to fully resolve the effects of the propeller during the ship turning. The limitation of the simplified actuator disk model may lead to a difference in the roll response from the experimental results.

The calculated time histories of the critical values in head waves showed qualitatively similar trends to those estimated by Wang and Wan (2018) for the ONR Tumblehome ship in head waves (length ratio $\lambda/L_{BP} = 1.0$ and wave steepness $H/\lambda = 0.02$). Their findings revealed that the ONR Tumblehome ship also experienced a significant speed loss which can be as large as 40% during the turning circle manoeuvre. Also, the maximum pitch and heave motions of the ONRT were observed to be $2.5°$ and 0.035m, respectively. These motion values are approximately 14% larger than those of the KCS in this work, which may be due to the difference in the applied wave height for the turning circle manoeuvre.

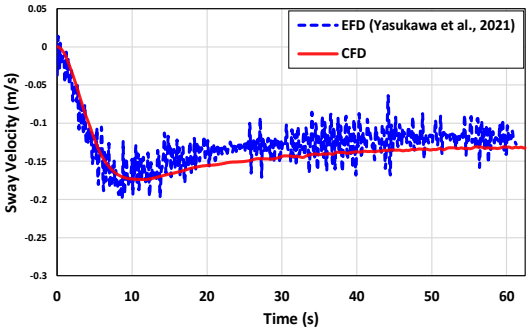
Regarding the values related to the propeller disk model in Fig. 13 and Fig. 14 (g - h), the thrust and torque were underestimated to some extent, while the overall trend matched quite well with the experimental data. The calculated steady thrust and torque of the disk model during the turning manoeuvre in calm water were underestimated by 5.98% and 11.57%, respectively. Similarly, the values in head wave were underestimated by 5.42% and 12.5%. It should be noted that the propeller is under the influence of cross flow, waves, and wave-induced ship motion during the ship manoeuvres. However, the actuator disk based on the body force method may not be capable of fully resolving these critical aspects. The present disk model also did not consider the side force of the propeller, as discussed in Dubbioso et al. (2017). Therefore, it is thought that these physical limitations might cause a difference in thrust and torque from the experiments.

Fig. 15 (a) and (b) describe the turning trajectories in a calm and head sea, respectively. The CFD results of turning circle trajectories for both cases are in reasonable agreement with the measurements. The calculated turning trajectory in calm water was slightly larger than that of the experimental data, whereas the trajectory in head wave was slightly smaller than that of the measurements. Despite these different patterns of the turning trajectories in both cases, the important turning indices related to the ship turning manoeuvre were estimated fairly well with a maximum error of about 8% using the current CFD approach as can be seen from the results listed in Table 7 and Table 8. This comparison indicates that the present numerical approach can provide an overall assessment of the manoeuvring performance in both calm sea and waves.

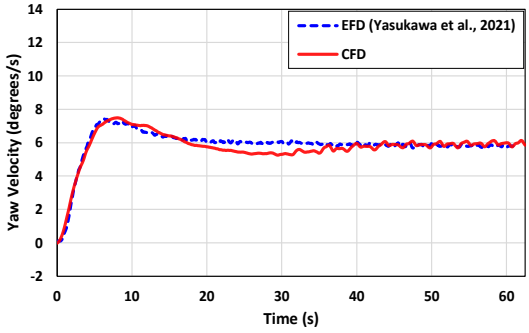
With this comparison, the current CFD approach can be argued to be validated and can be applied for further examinations. This model was therefore used throughout all the simulations.



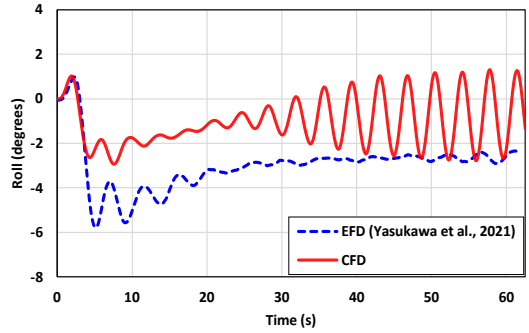
(a) Surge velocity



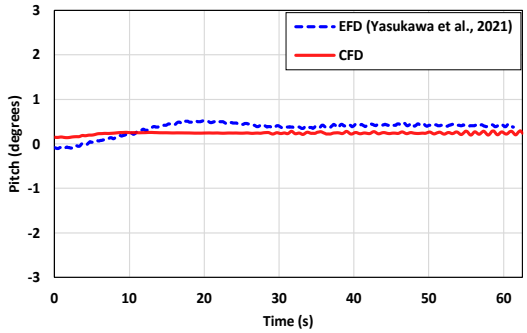
(b) Sway velocity



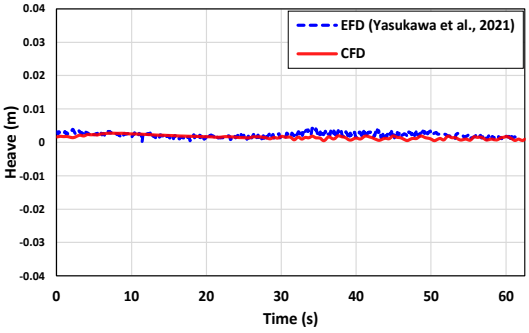
(c) Yaw velocity



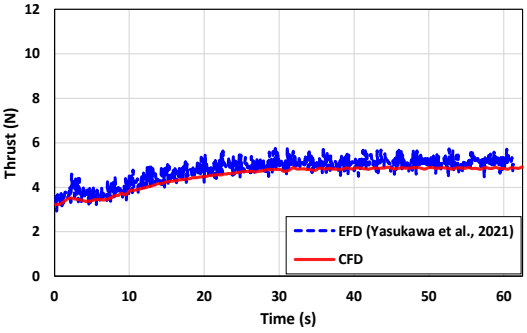
(d) Roll displacement



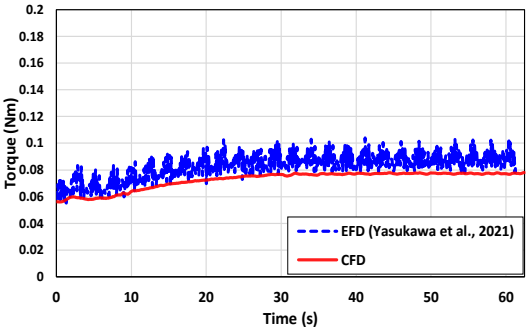
(e) Pitch displacement



(f) Heave displacement



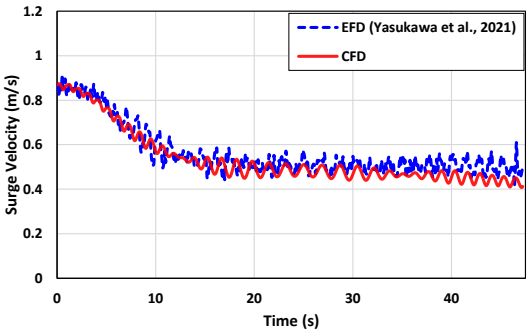
(g) Thrust



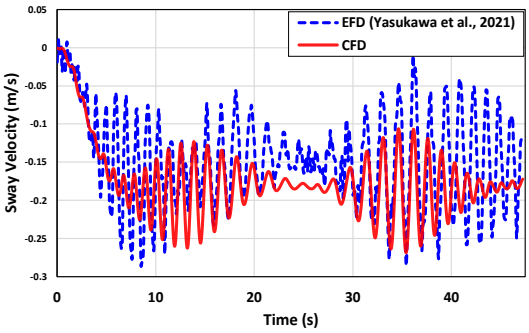
(h) Torque

Fig. 13 Comparisons of the time histories of the ship velocities, motions, and propeller characteristics during

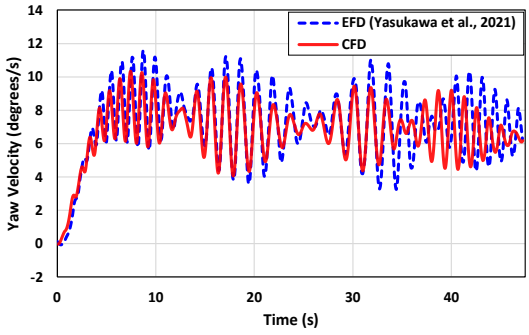
turning a circle manoeuvre in calm water.



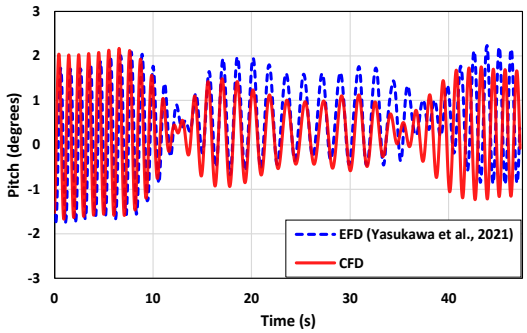
(a) Surge velocity



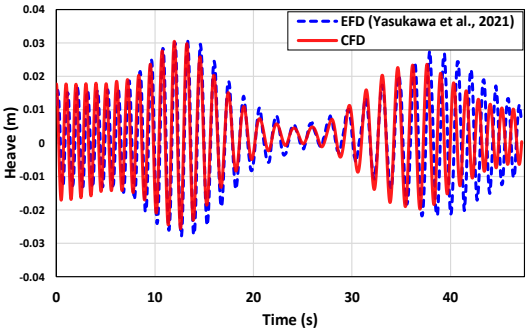
(b) Sway velocity



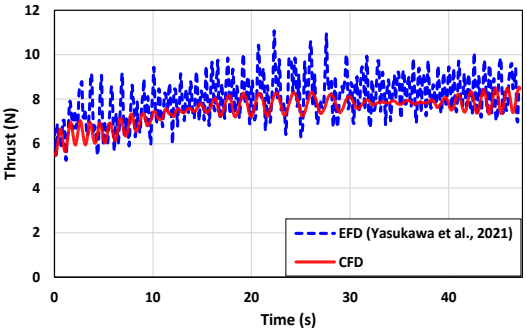
(c) Yaw velocity



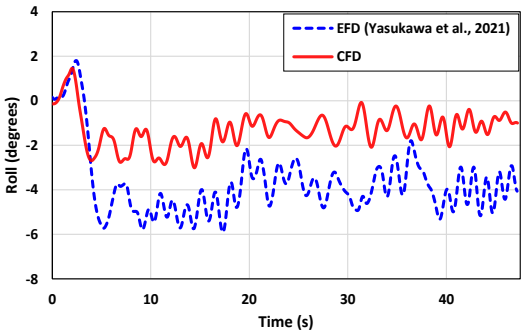
(e) Pitch displacement



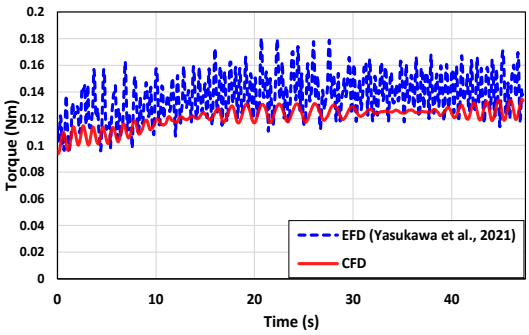
(f) Heave displacement



(g) Thrust

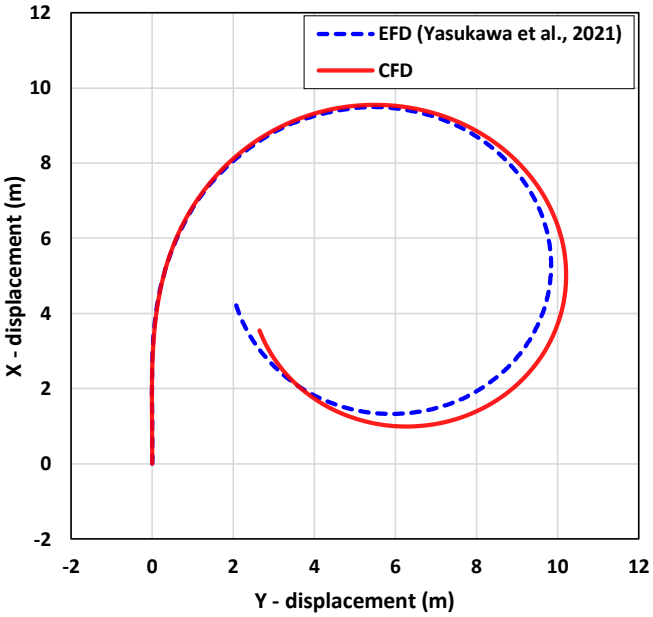


(d) Roll displacement

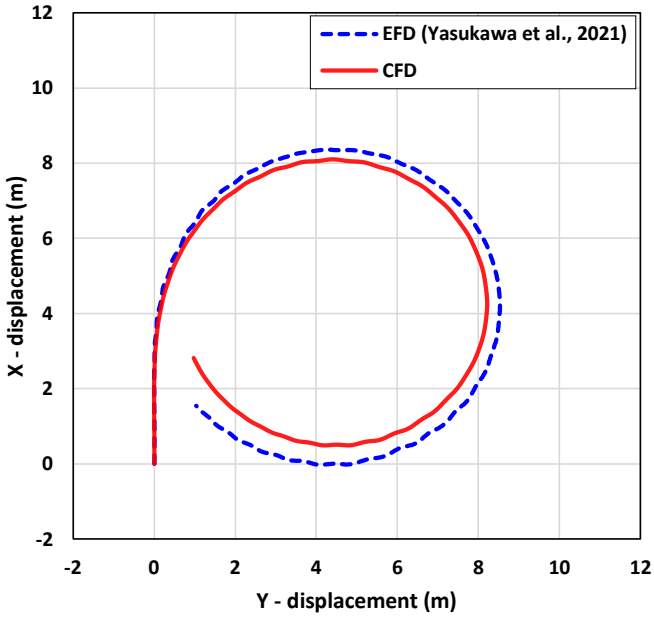


(h) Torque

Fig. 14 Comparisons of time histories of ship velocities, motions, and propeller characteristics during a turning circle manoeuvre in head wave.



(a) Trajectory in calm water



(b) Trajectory in head waves

Fig. 15 Comparisons of turning trajectories in a) calm water and b) head waves.

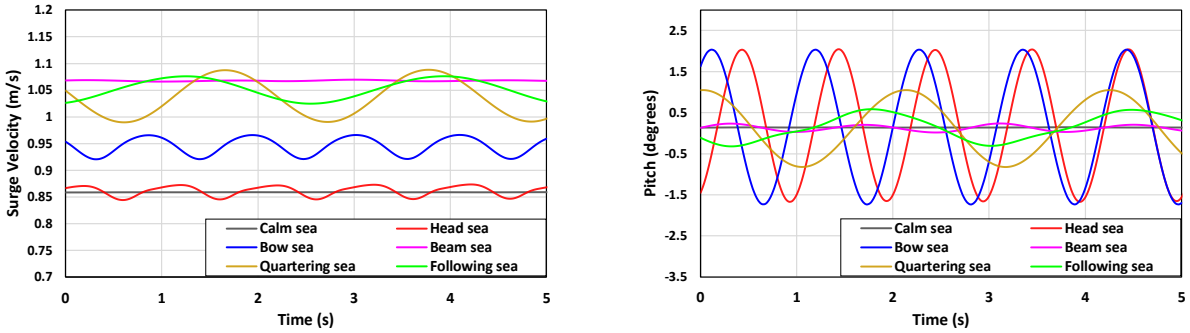
5.4. Self-propulsion

The self-propulsion condition is a required procedure which should be accomplished before starting the ship turning manoeuvre. The results obtained from the self-propulsion computations can provide an investigation into the manoeuvring performance of a ship under the same propeller revolution for different wave conditions. It is worth noting that the self-propulsion tests were conducted with 6 degrees of freedom, with the moving rudder controlled by the feedback controller to maintain the ship's course before starting the turning manoeuvre. As listed in Table 2, the nominal approach speed for both calm and head sea conditions (case 1 and 2) was set to be 0.86 m/s with a propeller speed of 10.56 RPS and 13.38 RPS, respectively. For the rest of the simulations (case 3 - 6), the speed of the propeller at self-propulsion was set to be 13.38 RPS, which equals to the revolution of the head wave case.

Fig. 16 shows the comparisons of the time histories of the approach speed, ship resistance, pitch motion, and heave motion under the self-propulsion conditions for all the cases. The Fourier Series (FS) (Tezdogan et al. 2015; Wang and Wan 2018) was utilised to analyse the harmonic CFD results due to wave motion. The average value of the time history of the obtained results was calculated by 0th harmonic FS term and the mean amplitude of the oscillation of the values was quantified by the 1st harmonic FS term. The obtained FS results are given in Table 9.

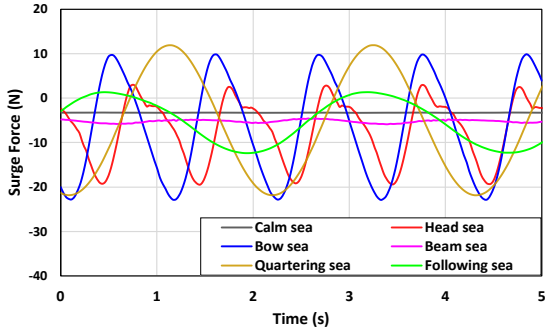
As Fig. 16 and Table 9 jointly show, the propeller speed was increased by 27% in head waves compared with in calm water in order to achieve the approach speed (0.86 m/s), while the total resistance also increased by 226% due to the incident head waves. As stated previously, regular waves with the length ratio of $\lambda/L_{BP} = 1.0$ and height of $H = 0.048\,m$ were applied throughout all the simulations. When the ship was advancing forward in waves, she experienced substantial changes in the forward speed and resistance. The increase of the mean forward speed was predicted to be 9.9% for the bow wave, 24% for the beam wave, 21% for the quartering wave, and 22% for the following wave when compared to the head wave case. Similarly, the decrease in ship resistance was found to be 6% for the bow wave, 30% for the beam wave, 27% for the quartering wave, and 29% for the following wave. From this observation, the ship operation in head wave can lead to a drastic increase in the fuel consumption.

The comparisons between the time histories of the ship motions in various wave conditions are presented in Fig. 16(c) and (d) to better understand the effects of wave directions on the ship motions. It should be reiterated that the frequency of ship motions in waves is determined by the wave encountering frequency. Short-term oscillations of the ship motions with regular periodicity were observed in head waves whereas relatively long-term oscillations were found in following waves within a given period due to the wave encounter frequency. The amplitude of ship motions was found to be closely related to the wave direction. The pitch motion was considerably increased in head and bow seas with a mean amplitude of 1.84º and 1.87º, whilst the pitch in beam waves was almost negligible with a value of 0.4º. Also, the decrease in the amplitude of pitch oscillation was predicted to be 95% for the beam sea, 49% for the quartering sea, and 78% for the following wave, compared with the head wave case. As for the heave motion, the largest oscillation amplitude was predicted in beam waves with a value of 0.025m which is almost equal to that of the incident wave: only a difference of 4%. Also, the increase in the heave amplitude was found to be 12% for the bow sea and 27% for the beam sea, while the reduction was estimated to be 53% for the quartering sea and 94% for the following sea, compared to the head wave case.

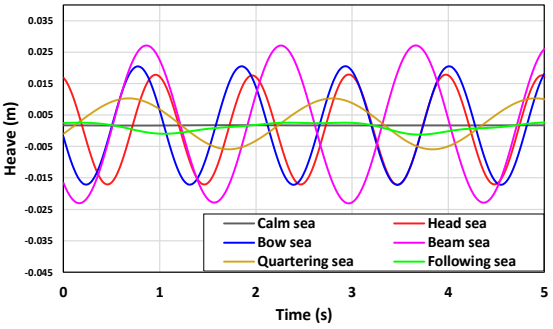


(a) Approach speed (U_0)

(c) Pitch displacement



(b) Surge Force



(d) Heave displacement

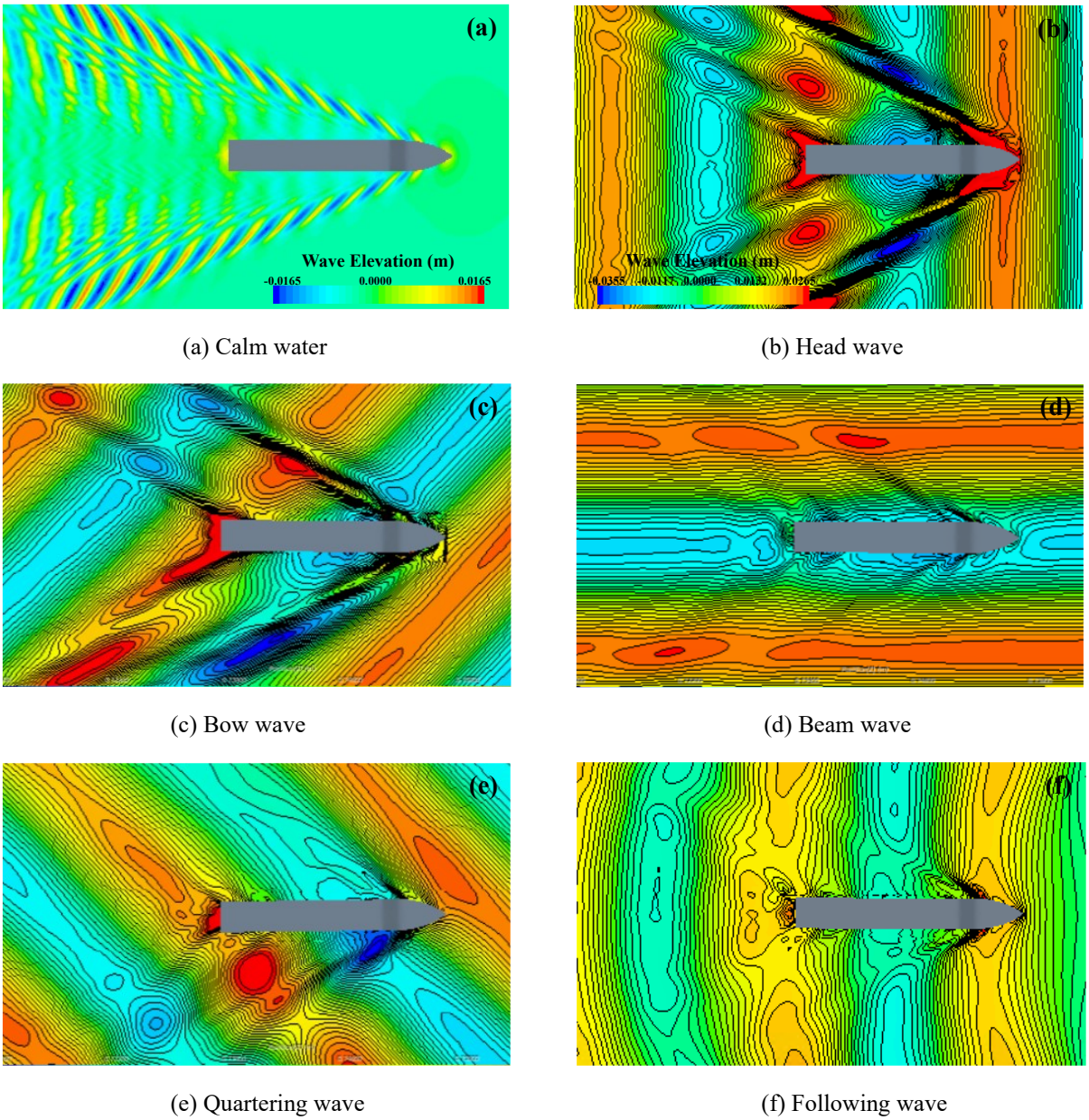
Fig. 16 Time histories of the approach speed, ship resistance, pitch motion and heave motion for all cases.

Table 9 Fourier Series analysis of the approach speed, ship resistance, pitch motion and heave motion at self-propulsion.

Case no.	Approach speed U_0 (m/s)		Resistance F_x (N)		Pitch displacement (degrees)		Heave displacement (m)	
C	0th FS term	1st FS term	0th FS term	1st FS term	0th FS term	1st FS term	0th FS term	1st FS term
1 (Calm sea)	0.860	-	3.237	-	0.002	-	0.144	-
2 (Head sea)	0.860	0.013	7.318	9.240	0.180	1.842	0.000	0.017
3 (Bow sea)	0.945	0.024	6.723	15.548	0.170	1.876	0.002	0.019

4 (Beam sea)	1.067	0.008	5.186	0.438	0.123	0.095	0.002	0.025
5 (Quartering sea)	1.038	0.048	5.336	16.843	0.127	0.931	0.002	0.008
6 (Following sea)	1.049	0.025	5.215	6.619	0.135	0.399	0.001	0.001

The instantaneous free surface elevation at the self-propulsion condition is shown in Fig. 17 in which the Kelvin waves generated by the advancing ship are clearly visible. In the head wave case (b), the rudder angle was deflected towards starboard 35° during the moments when the wave trough passed on the midship of the model based on the experimental data. In the same way, for the rest of the simulations in waves, the rudder started to be deflected during the moments when the wave trough passed on the midship of the ship. However, it is worth noting that Kim et al (2019) stated that the timing of rudder deflection has little influence on the low frequency manoeuvring characteristics of a ship.



(a) Calm water

(b) Head wave

(c) Bow wave

(d) Beam wave

(e) Quartering wave

(f) Following wave

Fig. 17 Free surface elevation at self-propulsion condition before conducting the turning manoeuvre.

5.5. Course keeping control

In general, a safe navigational route should be determined before a voyage commences. A vessel should navigate according to the prescribed route on the sea to ensure a safe navigation. The rudder is controlled based on the PID or adaptive control built into the auto-pilot in order to follow the planned route on the sea states. Deviation from the original course occurs frequently in waves, due to the difficulty of controlling the ship's heading under the external wave forces, which may lead to an increase in the sailing distance and hence ship operating costs. More importantly, any deviation from the planned course is likely to cause serious maritime accidents like ship grounding due to the change in the water depth, from deep to shallower. Thus, it is of great importance to understand the course keeping behaviour of a vessel. With this present context, IMO (2017) has also been focusing on course keeping ability by introducing guidelines relating to the steering ability of a ship in adverse weather to ensure the safety of the navigation.

The key advantage of the present CFD approach is that it can predict not only the combined seakeeping and manoeuvring performance but also the course keeping ability under different wave conditions. Wang et al (2017), for example, carried out direct CFD simulations successfully to evaluate the course keeping ability of an ONR Tumblehome ship in waves using the fully nonlinear unsteady RANS method. Unfortunately, to the best of our knowledge, no well-documented experimental tests for the course keeping behaviour of the KCS model in waves are available. Though, as presented in sub-section 5.2, we can observe good agreements for the yaw velocity according to the rudder deflection, which is the most influential factor for the course keeping control. Thus, it is considered to be acceptable to investigate the course keeping ability in waves by using the current CFD model.

Fig. 18 and Fig. 19 illustrate the time evolution of the rudder, yaw angle, and yaw velocity at the self-propulsion condition for all simulation cases. Based on the applied coordinates system, the positive value of the rudder angle indicates the rudder deflection for making the ship's heading to port, whereas the negative value indicates the deflection for making the ship's heading to starboard. A plus sign of yaw angle and velocity refers to the direction of rotation to starboard, whilst a minus sign refers to the direction of rotation to port. It should be noted that the simulated results for the course keeping ability showed a similar trend between the rudder angle and the yaw angle since the rudder controller depends largely on the yaw deviation defined in Eq. (8). However, it should be borne in mind that the maximum execution angle of the rudder can be changeable according to the coefficients of the PID controller.

As presented in Fig. 18 (a) and (b), the rudder control was not critical to maintain a desired heading of the ship in both calm water and head waves. As for the calm water case, the yaw angle was kept to be almost 0° with the rudder deflection about 1.8°. It is worth noting that an advancing ship has a small yaw moment caused by the asymmetric pressure distribution acting on the rudder blade, due to the discharge flow of the propeller, which can result in the small rudder deflection. Concerning the head wave simulation, the maximum execution angle of the rudder was noted to be about 0.8° with the small fluctuation. The maximum yaw velocity was also predicted to be 0.008°/s in calm water and 0.009°/s in head wave. As seen in Fig. 17 (a) and (b), the wave patterns are symmetric when the ship is advancing straight forward in both calm and head seas. This means that there is almost no pressure difference on both sides of the ship hull, which does not induce lateral force and yaw moments and thus does not induce yaw deviation.

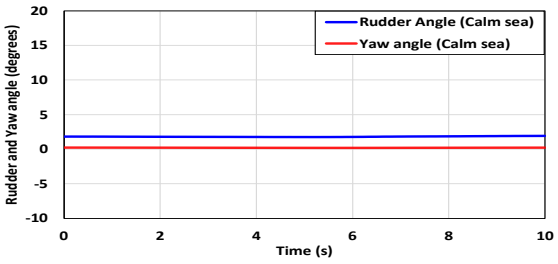
Compared with the calm water and head wave cases, the maximum rudder deflection became noticeably larger in beam and following seas. The rudder deflection in beam and following seas was observed to be 3.1° and 3.7°, respectively. The maximum yaw deviation was also reported to be 0.41° in beam sea and 0.46° in following wave, as presented in Fig. 18 (d) and (f). The maximum yaw velocity was about 0.3°/s for both cases. Similar to the calm water and head seas, symmetric wave profiles around the hull were found, which can barely affect the yaw deviation of the ship, as depicted in Fig. 17 (d) and (f).

On the other hand, the rudder angle deflection becomes large to make the ship straight in oblique seas, namely bow and quartering wave. As presented in Fig. 17 (c) and (d), asymmetric wave profiles were observed in oblique seas, which may cause a substantial lateral force and yaw moment. The maximum rudder deflection was observed

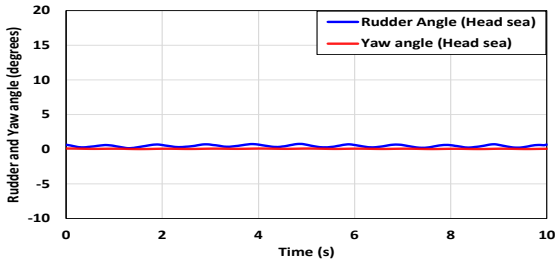
to be 6.8° in bow sea and 12° in quartering sea, while the maximum yaw velocity was 2.4°/s and 3.4°/s, respectively. It can be clearly noted that these values showed a larger amplitude of oscillations in the simulations presented in Fig. 18 (c) and (e) and Fig. 19 (c) and (e). The maximum value of yaw deviation was also calculated to be 0.6° and 1.4°, respectively. It appears that the oblique seas experienced the large rudder angle deflection based on the rudder feedback controller to control the ship's heading among all the wave simulations.

Fig. 20 illustrates the predicted ship path comparison under course keeping control for all the simulations. Most of the paths for the simulated cases in waves were clearly different from the trajectories obtained in the calm water condition. The trajectories can be affected by the complex interaction of hull-propeller-rudder under the incident waves. A very small deviation from the original course was noticed in both calm water and head seas, which can be believed to achieve good course-keeping control. The oscillating deviation of the trajectory from the original path was found in beam seas, which may be due to the strong lateral force induced by incident beam waves. It is of note that the increasing deviations from the original course were observed in bow, quartering and following sea, which indicate a relatively poor performance of course keeping control compared with other cases. The bow sea experienced the largest deviation from the original course, which is due to the strong lateral force and yaw moment caused by the oblique sea. With the same reason, the quartering wave also suffered a large deviation. The ship's course in the following sea was observed to be more biased towards the starboard side than the calm and head sea cases. This may be due to a larger pressure difference acting on the rudder surface as a result of the interaction between the wave and the propeller-induced flow, which makes the ship's heading more towards the starboard side. From the predicted CFD results, it can be derived that the course keeping performance of a ship in a rough head sea can be a very challenging task.

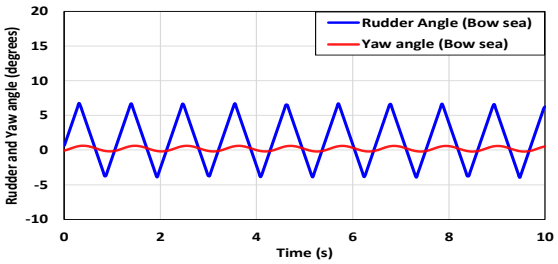
In navigation practice, there exists a certain delay from the time when the rudder angle is given by the controller to the moment the response of the steering engine is completed (Yan et al., 2020). However, the feedback controller applied to the present CFD model did not consider the steering delay, which may result in some discrepancy between the CFD results and the actual rudder response. For example, Wang et al. (2017) also conducted the course keeping manoeuvre in waves using the feedback controller without the consideration of the steering delay. The comparison between the predicted rudder response and the measurements showed some discrepancy in terms of the rudder deflection frequency. Given this, it should be noted that the steering delay is also an important factor for the course keeping control.



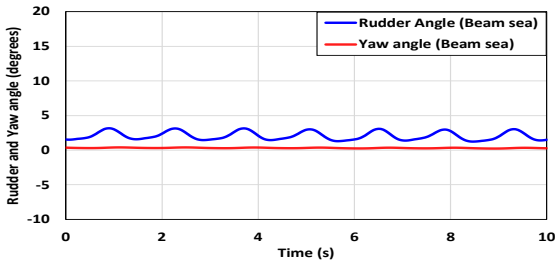
(a) Rudder and Yaw angle in Calm sea



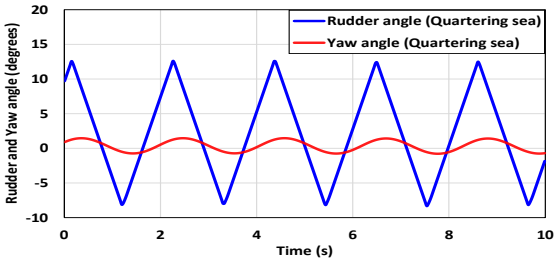
(b) Rudder and Yaw angle in Head sea



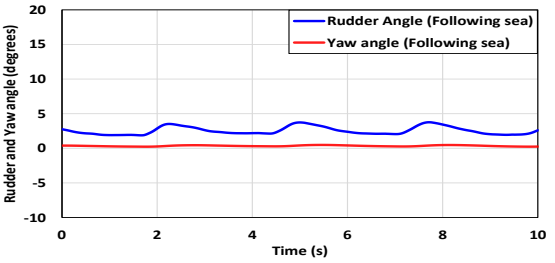
(c) Rudder and Yaw angle in Bow sea



(d) Rudder and Yaw angle in Beam sea

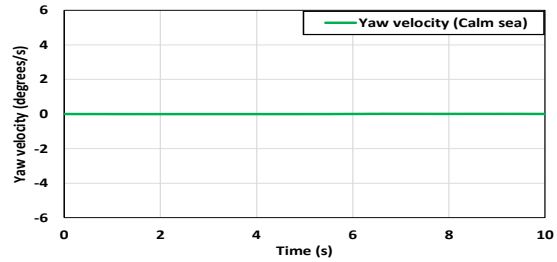


(e) Rudder and Yaw angle in Quartering sea

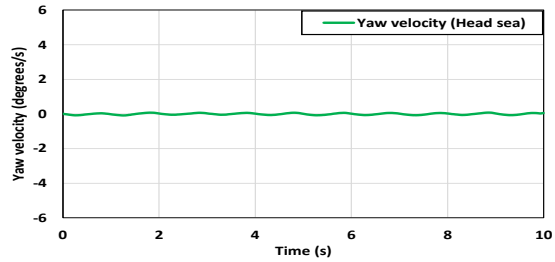


(f) Rudder and Yaw angle in Following sea

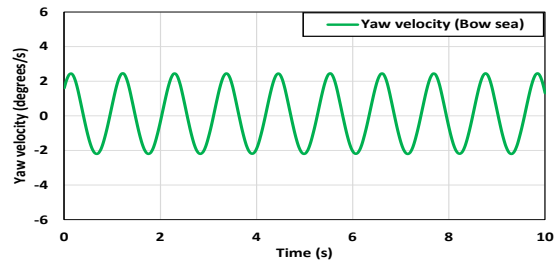
Fig. 18 Time histories of the rudder deflection and yaw angle at self-propulsion in (a) calm sea, (b) head sea, (c) bow sea, (d) beam sea, (e) quartering sea, and (f) following sea.



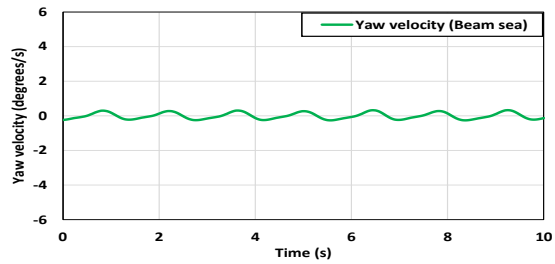
(a) Yaw velocity in Calm sea



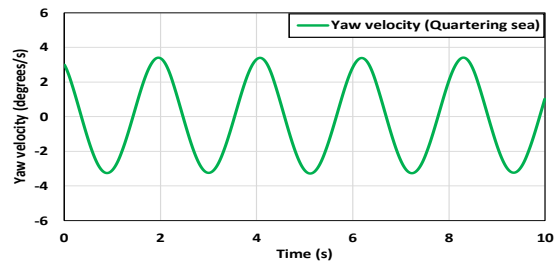
(b) Yaw velocity in Head sea



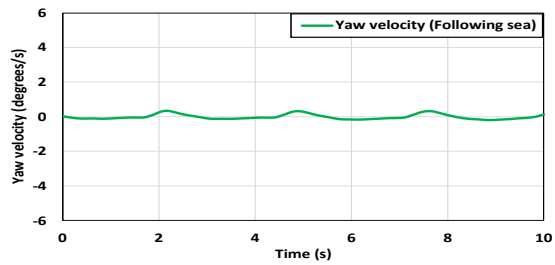
(c) Yaw velocity in Bow sea



(d) Yaw velocity in Beam sea



(e) Yaw velocity in Quartering sea



(f) Yaw velocity in Following sea

Fig. 19 Time histories of yaw velocity at self-propulsion in (a) calm sea, (b) head sea, (c) bow sea, (d) beam sea, (e) quartering sea, and (f) following sea.

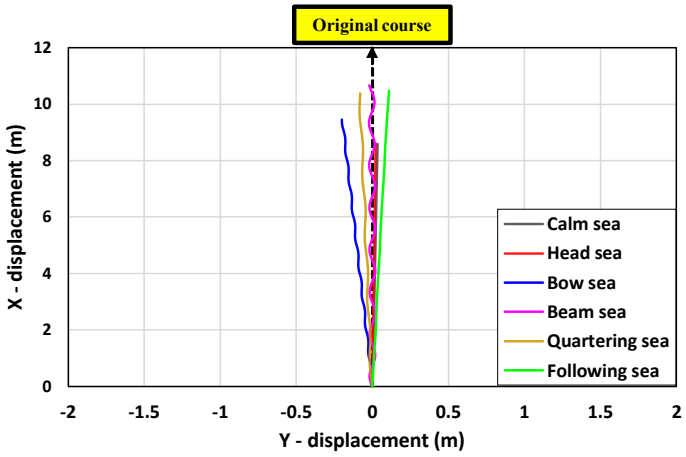


Fig. 20 Comparison of the predicted trajectories in waves.

5.6. Turning circle manoeuvre in waves

This sub-section outlines direct CFD calculations of a standard turning manoeuvre in waves. For all manoeuvre simulations in waves, the ship is moving forward at a constant speed and the rudder is executed to a maximum rudder angle of 35° in the starboard direction at maximum rudder rate. The ship responds turning towards the starboard side. The manoeuvre is completed when the 360° turn is achieved.

Throughout all the simulations, the turning circle test with only the yaw angle variation up to 360° was carried out according to the general procedure (ITTC, 2002). A 720° turn is also recommended in order to fully assess environmental effects, but this requires a huge computation time and hence high computational resources. Thus, in this study, a 720° turn was not investigated, assuming that the 360° turn provides sufficiently valuable information about the manoeuvring characteristics of a ship in waves.

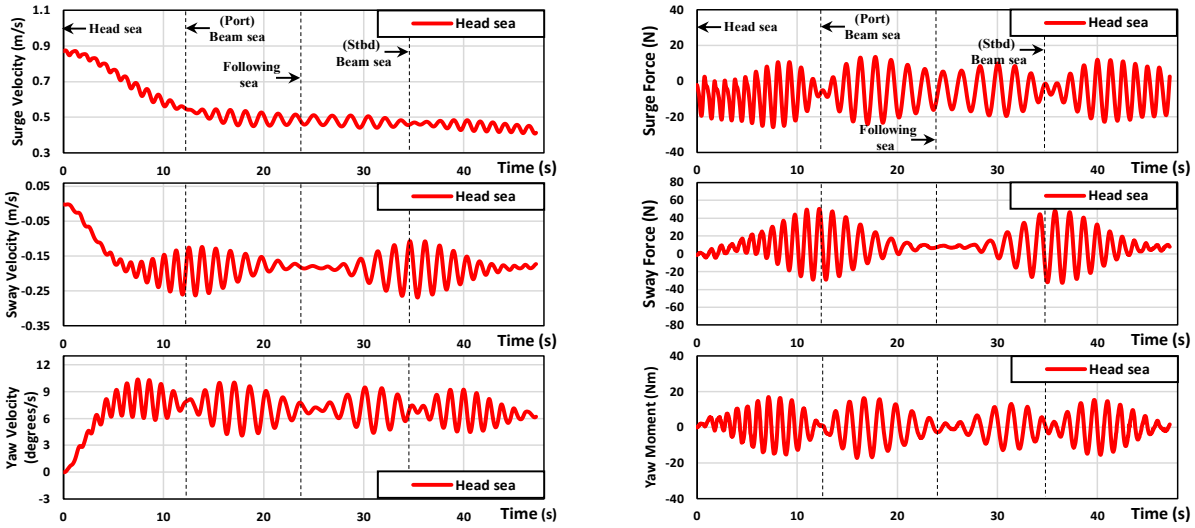
5.6.1. Turning indices in waves

The obtained time histories of the ship velocities, forces, and moment during the turning circle manoeuvre in waves are illustrated in Fig. 21, in which the representative wave-encounter directions are marked with black dashed-lines. Unlike other hydrodynamic problems such as ship resistance and seakeeping, the present manoeuvre simulations experience instantaneous variations of wave-encounter conditions during the ship's turning. Such instantaneous variations make the manoeuvring simulations in waves more complex and therefore leads to high computational costs. All values on the graphs are calculated for the model-scale ship geometry.

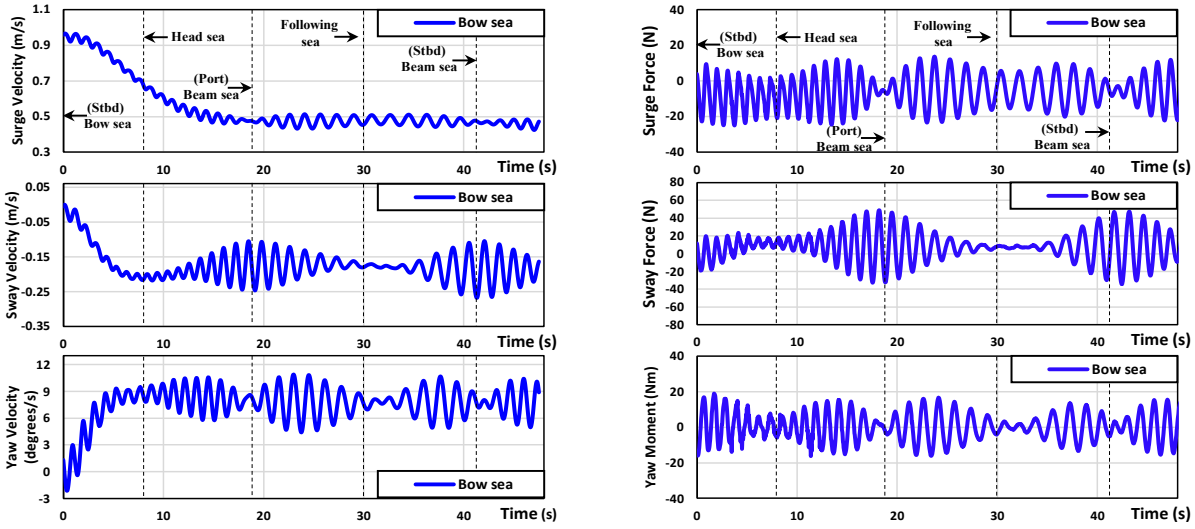
From Fig. 21, it can be observed that the ship forward speeds in all cases gradually decreased after steering and almost converged to a similar value slightly below 0.5 m/s with some fluctuations around their mean value. This involuntary speed loss is correlated with a large drift angle, which causes an increase in ship resistance. The maximum speed loss in all wave cases occurred between approximately 20s – 25s after the rudder was deflected. After this, a slight increase in forward speed was seen when the ship experienced beam, quartering, and following waves, whereas a slight decrease was observed in head and bow waves. Because of this, the time histories of the forward speed in waves became rather similar despite the large differences in the approach speeds at an early stage of the turning manoeuvre. The predicted speed loss between the initial approach speed and the steady mean value was 48% in head sea, 51% in bow sea, 56% in beam sea, 54% in quartering sea, and 55% in following sea. It is of note that the surge forces were largely oscillated during the turning manoeuvre in waves, but the oscillations significantly decreased at the moment when the ship experienced the incident wave from port or starboard beam. The surge velocities also followed the same trend.

When it comes to the sway velocities and forces, they showed rapid increases until about 10s after the rudder was deflected to starboard and then gradually converged to similar values with some oscillations induced by the

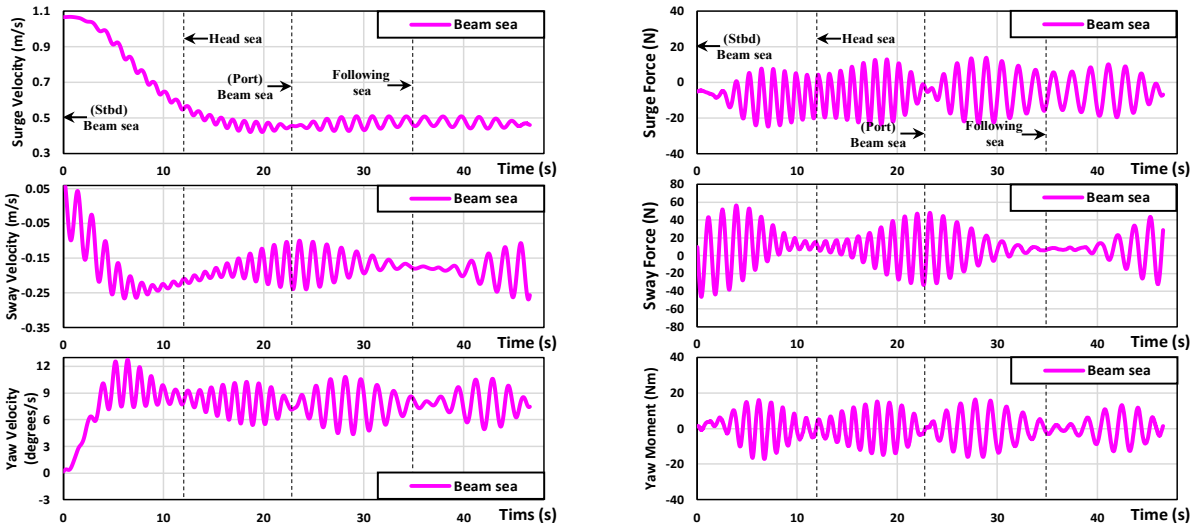
waves. The converged mean values of the sway velocities were reported to be approximately -0.18 m/s in head sea, -0.19 m/s in bow sea, -0.18 m/s in beam sea, -0.17 m/s in quartering sea, and -0.16 m/s in following sea. The largest fluctuations in the sway velocities and forces were also observed at the moment when the ship encountered beam seas, whereas the smallest fluctuations were found under both the head and following sea conditions. As for the yaw velocities and moments, relatively large fluctuations clearly occurred when the ship encountered oblique waves. The yaw rates were maximum, occurring about 7s after the rudder deflection. Then, they converged quickly to the values which were estimated to be 6.60 °/s in head sea, 7.82 °/s in bow sea, 7.84 °/s in beam, 8.04 °/s in quartering sea, and 7.35 °/s in following sea.



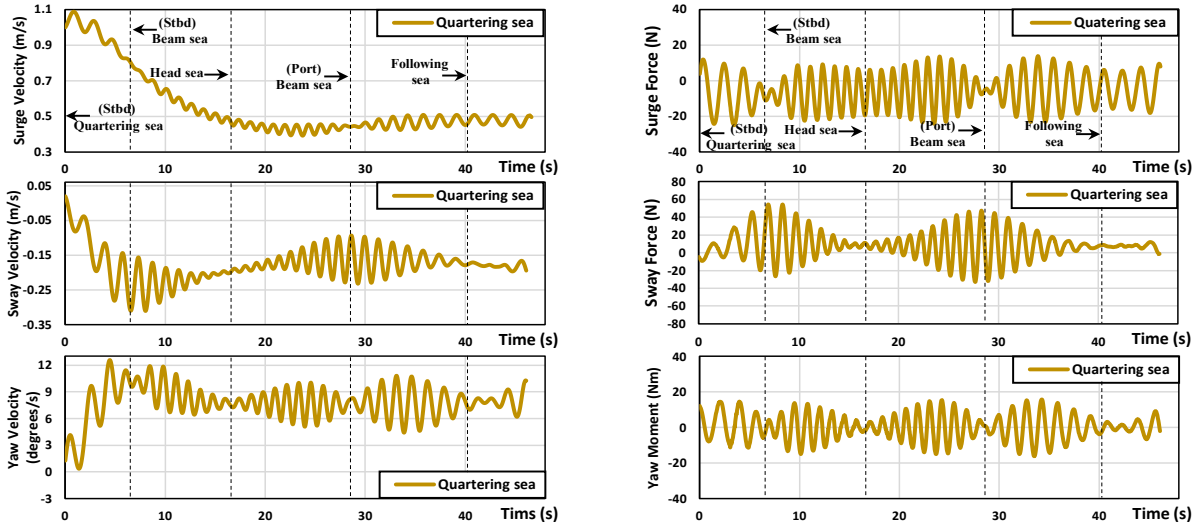
(a) Time histories of head wave case



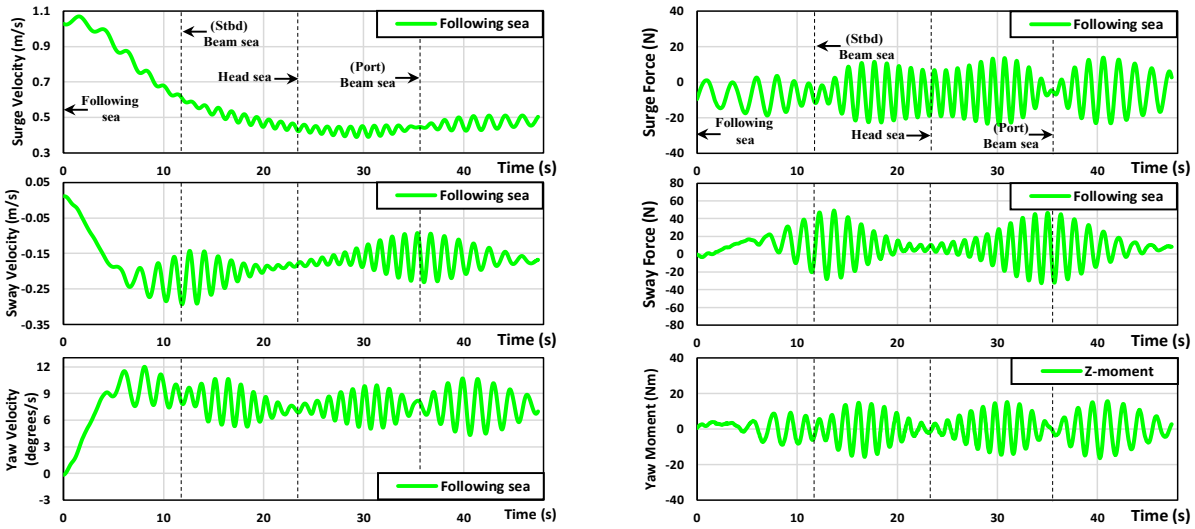
(b) Time histories of bow wave case



(c) Time histories of beam wave case



(d) Time histories of quartering wave case



(e) Time histories of following wave case

Fig. 21 Time histories of the ship velocities, forces, and moment during a turning manoeuvre in waves.

Table 10 describes the turning manoeuvring indices under various wave conditions. The critical manoeuvring parameters were closely associated with the ship's horizontal motions during the turning manoeuvre. Thus, the hydrodynamic forces and moment acting on the ship in the horizontal plane were identified as the most important factors from the manoeuvring point of view: the surge force (added resistance), sway force, and yaw moment (Skejic and Faltinsen, 2008). During the ship's turning, the resultant forces and moment were induced by complex interactions between the rudder, propeller, hull, and waves. In particular, the imposed incident wave conditions had a great influence on the hydrodynamic forces and moment. As a result, there was a high probability that the turning circle trajectory would be deformed. The ship advance would be closely correlated to the ship velocities in the horizontal plane (surge, sway, and yaw velocities). In general, the greater the forward speed and the longer time for the yaw angle to change by 90º, the larger the ship advance can be. Although the 90º turn time was relatively long, the minimum advance was observed in the head sea (case 2) because the forward speed was the lowest at the beginning of the turning manoeuvre. The advance in the bow sea (case 3) was far greater than that of the head sea case (case 2) due to the relatively large forward speed and the long 90° turning time. It was noted that the beam (case 4), quartering (case 5), and following wave (case 6) had caused a larger yaw moment than that in the head and bow waves as the ship started turning in waves. This accelerated the ship's yaw velocities, and 90° turns were achieved much earlier under the beam, quartering, and following seas. The ship advances were predicted larger than the head sea case due to the large forward speeds despite the short time to 90º turn in the beam, quartering, and following seas.

The ship transfer was also found to have a close correlation with the surge, sway, and yaw velocities. The larger the surge and sway velocity and the longer time to turn by 90º may lead to the greater transfer. The relatively small transfer was predicted in the head and bow wave cases due to the relatively small surge and sway velocities, despite the long time to turn by 90º. In the early stage of the turning, it is of note that the lateral forces in the beam, quartering, and following wave appeared to be larger than those observed in the head and bow waves. Such a large sway force accelerated the ship's sway velocities, and relatively large sway velocities were achieved under the beam, quartering, and following waves. Although the time taken for the 90º turn was short, the ship transfers were observed greater than in the head and bow sea cases due to the large surge and sway velocities. The tactical diameters were also clearly found dependent on the direction of waves. It was revealed that the relatively large tactical diameters were observed in the beam, quartering, and following sea cases under which a large lateral force and yaw moment were observed at the early stage of the turning.

The turning ability of the ship is considered satisfactory if the ship advance does not exceed 4.5 ship lengths (L_{BP}) and the tactical diameter does not exceed 5.0 ship lengths in the turning circle manoeuvre, according to the standards for ship manoeuvrability (IMO, 2002). All the obtained results of the turning manoeuvre were evaluated

by the IMO criteria in Fig. 22. From this comparison, it is seen that the turning ability of the KCS in waves can be regarded as satisfactory.

All the calculated turning trajectories of the KCS in waves are depicted in Fig. 23. It should be noted that all the rudder execution points were shifted to the specific location (0,0) in order to provide the correct comparisons of the turning trajectories. It is evident that some oscillations were estimated when the ship experienced starboard or port beam waves, which may cause large fluctuations of the sway velocities and forces.

Table 10 CFD results: turning indices in regular waves.

Parameters (CFD results)	Head sea (case 2)	Bow sea (case 3)	Beam sea (case 4)	Quartering sea (case 5)	Following sea (case 6)
Advance (m)	7.86 **$(2.57L_{BP})$**	8.85 **$(2.90L_{BP})$**	9.10 **$(2.98L_{BP})$**	8.44 **$(2.76L_{BP})$**	8.95 **$(2.81L_{BP})$**
Transfer (m)	3.09 **$(1.01L_{BP})$**	3.10 **$(1.01L_{BP})$**	3.43 **$(1.12L_{BP})$**	3.25 **$(1.06L_{BP})$**	3.56 **$(1.16L_{BP})$**
Time for yaw 90 degrees (s)	12.52	13.09	11.92	11.23	11.76
Tactical diameter (m)	7.97 **$(2.61L_{BP})$**	7.82 **$(2.56L_{BP})$**	8.17 **$(2.67L_{BP})$**	8.29 **$(2.68L_{BP})$**	8.86 **$(2.90L_{BP})$**
Time for yaw 180 degrees (s)	23.74	24.38	23.15	22.79	23.43

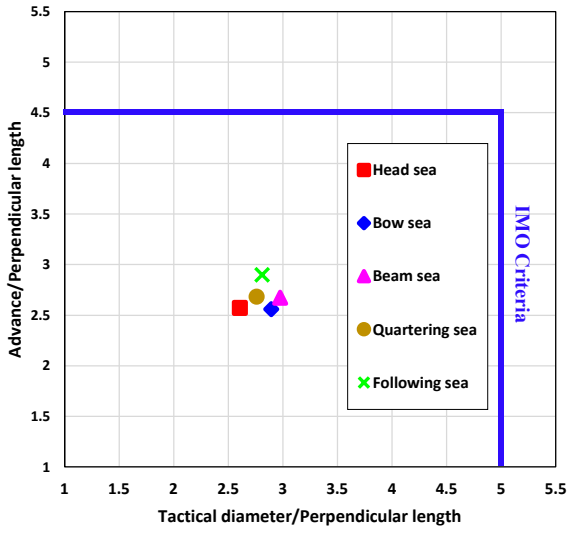


Fig. 22 Comparison of the predicted turning indices in waves with IMO turning criteria.

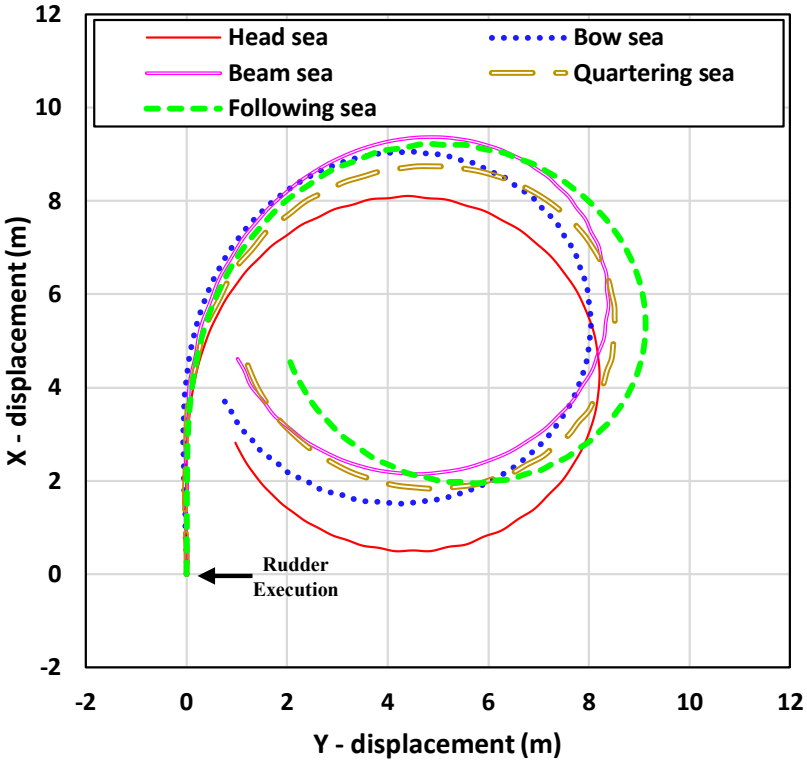
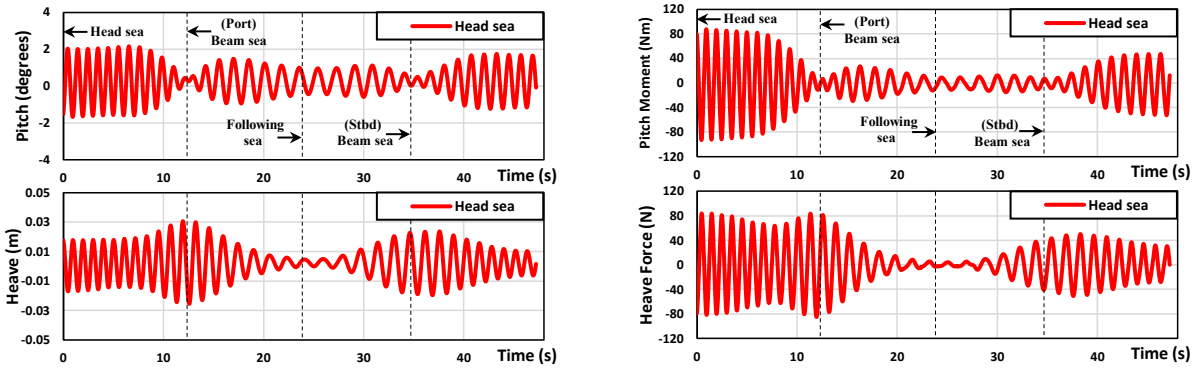


Fig. 23 Comparison of the predicted turning circle trajectories in waves.

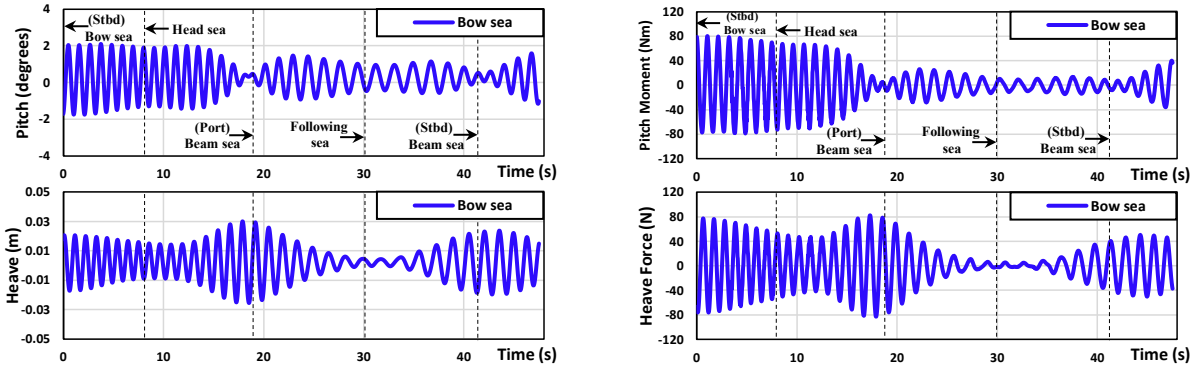
5.6.2. Wave-induced motions during turning

Fig. 24 shows the time evolutions of ship motions, i.e., heave and pitch, as well as pitch moment and heave force during the turning manoeuvre. As previously mentioned in sub-section 5.2, a large discrepancy was observed between the CFD and experiments so that this paper was not intended to discuss ship roll motions during the ship's turning in waves. In the figure, it can be clearly seen that the ship motions, force, and moment experienced considerable oscillations due to the high-frequency wave-induced motions. In addition, such frequencies continued to change during the turning manoeuvre due to instantaneous variations in the ship's velocity and wave-encounter direction. For example, a ship advancing in the head wave (case 2) encountered the head wave (0° turn), the port beam wave (90° turn), following wave (180° turn), starboard beam wave (270° turn), and head wave (360° turn) in series after starting the starboard turning manoeuvre. The ship's turning has contributed to continual changes in the wave-encounter frequency, consequently the ship motion frequency in waves.

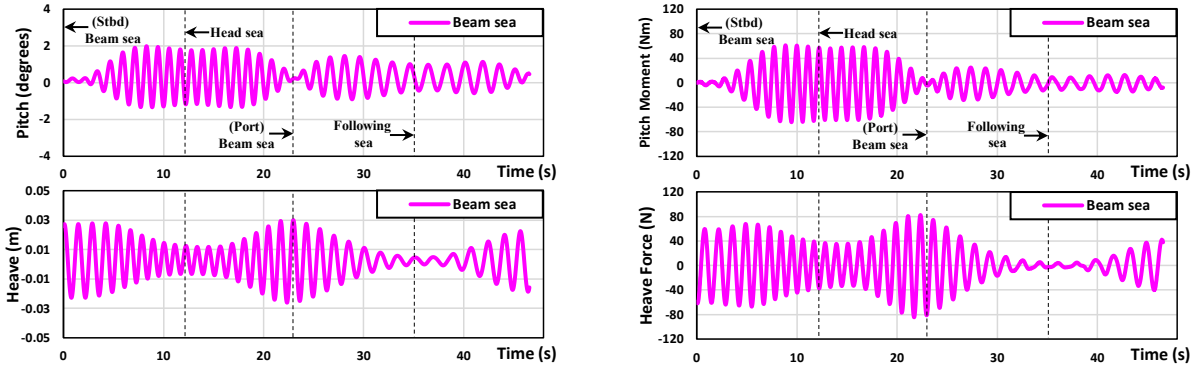
It can be seen from Fig. 24 that the wave-induced pitch motions showed significant variations in the amplitudes and frequencies of the motion during the turning circle manoeuvre according to the wave-encounter conditions. The maximum pitch motions were noted to be approximately 2º when the ship experienced the head or bow wave during the ship turning. On the other hand, the minimum pitch motions were almost 0º in the presence of the beam seas. The pitch moment also followed the same trend. In the same manner, the heave motion also experienced instantaneous variations of the amplitudes and frequencies of the vertical motion during the ship turning. The maximum heave displacement for all wave cases were predicted to be 0.03 m when encountering the port beam wave; on the other hand, the heave motion almost disappeared when the ship encountered following waves. It is of note that the heave force in the port beam wave was almost twice as large compared to that in the starboard beam wave during the ship turning, which directly affected the ship heave motion. The seakeeping results presented in Fig. 24 indicate that the ship motions showed qualitatively and quantitatively the same trend although phase differences were observed.



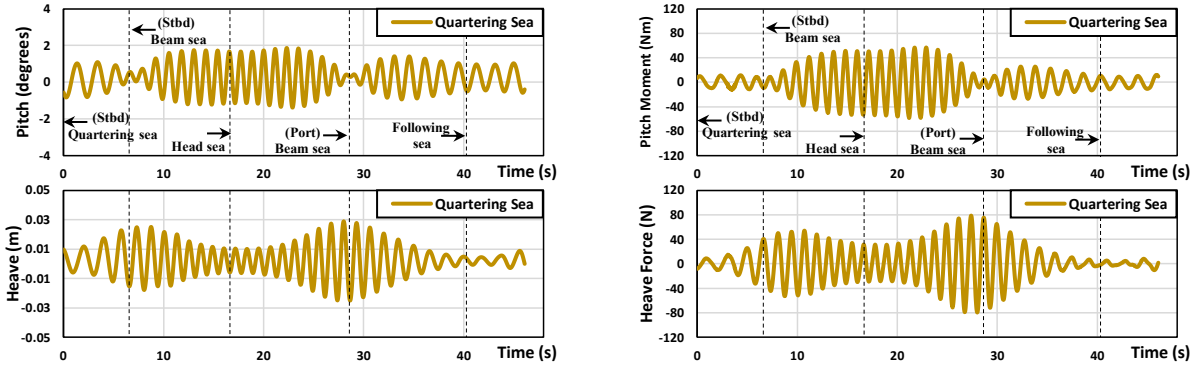
(a) Time histories of head wave case



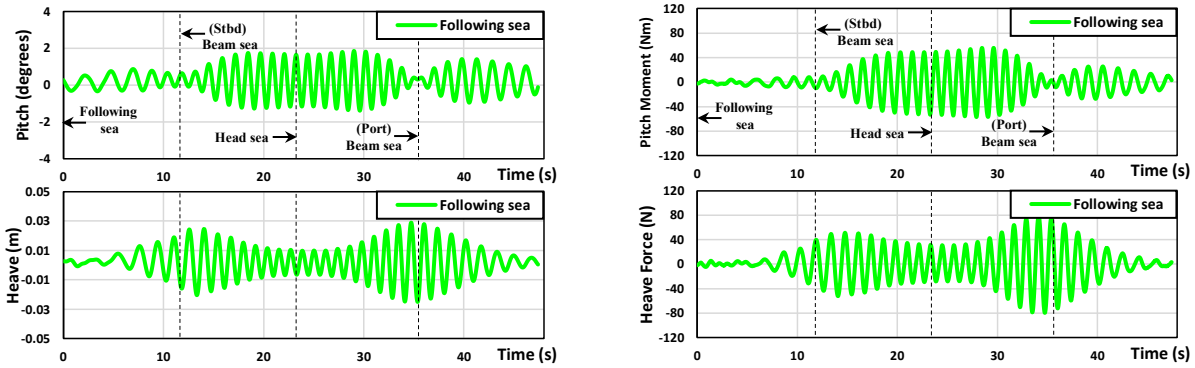
(b) Time histories of bow wave case



(c) Time histories of beam wave case



(d) Time histories of quartering wave case



(e) Time histories of following wave case

Fig. 24 Time histories of pitch, heave, pitch moment, and heave force during a turning manoeuvre in waves.

6. Concluding remarks

In the present study, direct CFD models for the prediction of the manoeuvring performance of a ship in waves have been developed. A model-scale KCS container ship appended with a turning rudder and a rotating actuator disk was used in our numerical simulations. The fifth order Stokes waves were adopted to generate the nonlinear regular waves. The manoeuvring analyses have been carried out using a commercial RANS solver, Star-CCM+.

A validation and verification study was carried out to quantify the numerical uncertainties of the present CFD models. Spatial and temporal convergence analyses were examined. It was shown in detail that the ship advance and transfer values, as well as the tactical diameter showed uncertainty levels of less than 0.2% for the ship manoeuvring simulation in head wave when calculated using the Grid Convergence Index (GCI) method.

For the validation of the present CFD models, the ship manoeuvring simulations in calm water and head waves were modelled using the current approach. It was demonstrated that the current CFD model predicted the turning indices, i.e., advance, transfer, tactical diameter as well as time to 90°/180° yaw angle change, within a range of -7.78% to 4.86% of the experimental data. For the time histories of the ship speeds, seakeeping motions, and propeller characteristics, the simulation results showed good agreement with the experiments. It is shown that the present CFD approach can be a reliable tool to assess a ship's manoeuvre in waves.

Fully nonlinear unsteady RANS simulations of the KCS were performed to directly simulate the ship manoeuvring characteristics in different waves, i.e., head wave, bow wave, beam wave, quartering wave, and following wave. From the obtained results, the effects of the wave travelling directions on the ship self-propulsion, course-keeping ability, and turning ability were investigated.

For the ship self-propulsion predictions, the approach speed under the same propeller revolution was calculated for each individual wave condition. From the CFD calculations it was observed that the minimum approach speed was found to be 0.86 m/s in head sea condition, whereas the maximum approach speed was noted to be 1.067 m/s in beam sea condition. This is due to the fact that the ship resistance significantly varies according to the incident wave directions. As for the seakeeping motions at self-propulsion condition, the head and bow seas considerably increased the pitch motion, which was almost negligible in the beam sea. However, the beam wave caused the largest oscillation amplitude of the heave motion.

For the course keeping manoeuvres in waves, five different directions were considered to evaluate the effects of wave directions on the course-keeping ability of the ship. For the head seas, the rudder control was found not critical to keep the ship straight. The maximum rudder angle was determined to be 0.8°, guaranteeing a good course-keeping ability. However, a relatively large deflection of the rudder angle was observed in oblique waves when the ship would keep moving forward. The rudder deflection was estimated up to approximately 7° in bow seas and 12° in quartering seas. This poor performance of course keeping control caused the large deviation from

the original course. It is concluded that the wave propagation directions can significantly affect the course-keeping ability of a ship, which consequently leads to deviations from the original course.

For the standard turning circle manoeuvres in waves, the predicted turning manoeuvring indices and trajectories in different waves were compared, from which it is concluded that the wave propagation directions have a considerable influence on the manoeuvring behaviour of a ship. The ship motions as well as the hydrodynamic forces and moments acting on a ship experienced considerable fluctuations during the turning manoeuvre due to the high-frequency wave-induced motions. The low-frequency responses caused by the ship manoeuvring motion were also observed. The free surface elevations around the ship were also explored. Also, it was observed that the Kelvin waves generated by the turning ship were more visible when the ship encountered head seas.

This study has provided a practical starting point for examination of a ship's manoeuvres in waves using CFD. Especially, the combined seakeeping and manoeuvring performance of a ship in different direction waves were observed. In addition to this, the impact of different wave conditions such as wave lengths and heights on the ship manoeuvre should be investigated. It has already been noted in real operations that a ship can experience diverse wave conditions in a real seaway. Therefore, a piece of future work may be the prediction of the ship manoeuvre in waves with different wavelengths and heights using the present direct CFD approach. This will help in improving understanding of a ship's manoeuvres in waves and hence improve navigational safety at sea. With the advance of computational resources, this CFD approach will also be expected to be an alternative tool to the conventional experiments investigating ship manoeuvring in waves.

Acknowledgements

It should be noted that all results were obtained using the ARCHIE-WeST High-Performance Computer (www.archie-west.ac.uk) based at the University of Strathclyde.

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